NUCES — FAST Islamabad

Department of Cybersecurity

TECHNICAL LABORATORY REPORT

Firewall Security Lab Assignment

FortiGate Next-Generation Firewall

Assignment No. 2 — Advanced Network Security

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Executive Summary

This technical laboratory report documents the end-to-end deployment, configuration, and validation of a FortiGate Next-Generation Firewall (NGFW) in a virtualized enterprise network environment. The assignment demonstrates practical competency in four core domains of network security engineering: firewall virtualization and deployment, secure management interface configuration, multi-layer policy implementation, and real-world threat validation.

The laboratory exercises were performed using FortiGate VM running under VMware Workstation. Four distinct tasks were executed: (1) VM deployment and interface provisioning, (2) HTTPS-based GUI management access hardening, (3) IPv4 firewall policy, Denial-of-Service (DoS) anomaly detection policy, and URL-based web content filtering policy implementation, and (4) live demonstration of website blocking and DoS attack detection.

Task	Objective	Status
Task 1	FortiGate VM Deployment & Interface Configuration	✓ Completed
Task 2	GUI Access Configuration & Admin Credential Hardening	✓ Completed
Task 3	Policy Implementation: IPv4, DoS, Web Filter	✓ Completed
Task 4	Live Demonstration: Blocking & Attack Detection	✓ Completed

All configurations were validated through CLI verification commands and GUI screenshots. Policy effectiveness was tested using Kali Linux as an attacker machine, executing hping3-based SYN flood attacks and browser-based URL access testing. Results confirm that the deployed FortiGate policies successfully mitigate the targeted threat vectors under laboratory conditions.

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1. Introduction

1.1 Background & Motivation

Network perimeter security forms the first line of defence in any enterprise security architecture. Firewalls act as gatekeepers, enforcing access control policies between trusted internal networks and untrusted external environments. FortiGate, developed by Fortinet, is one of the most widely deployed commercial NGFW platforms globally and consistently features in Gartner's Magic Quadrant for Network Firewalls.

This laboratory assignment provides hands-on experience with FortiGate's deployment lifecycle within a virtualised environment, bridging the gap between theoretical knowledge and operational practice required of cybersecurity professionals.

1.2 Laboratory Objectives

- Deploy FortiGate VM on VMware Workstation with correct virtual network adapter configuration
- Configure and verify the management interface for remote HTTPS-based GUI access
- Implement a baseline IPv4 firewall policy permitting outbound internet traffic with NAT
- Deploy an IPv4 DoS anomaly detection policy with threshold-based blocking
- Configure URL-based web content filtering to restrict access to social media (Facebook)
- Validate all deployed policies through practical live testing and screenshot evidence

1.3 Laboratory Environment

Component	Specification
Hypervisor	VMware Workstation Pro
Firewall Platform	FortiGate VM (FortiOS) — FortiGate-VM64.ovf
Firewall Resources	2 GB RAM, 30 GB HDD, 1 vCPU, 6 Virtual Network Adapters
Attacker Machine	Kali Linux (hping3 for DoS simulation)
Management Network	192.168.139.0/24 (VMNet — Host-Only)
LAN Network	Internal network — private IP range
WAN / Internet Interface	Port2 — ISP-facing interface with public IP
GUI Access Protocol	HTTPS (TCP/443)

2. Task 1 — FortiGate Firewall Deployment

This task covers the complete deployment pipeline of FortiGate VM within VMware Workstation: obtaining the VM image, provisioning virtual network adapters, importing the OVF template, and initialising the management interface via CLI.

2.1 Step 1 — Download FortiGate VM Image

The FortiGate VM image is obtained from the official Fortinet support portal. A registered account with a valid support contract or trial licence is required to access the downloads section.

Download URL: https://support.fortinet.com/support/#/downloads/vm

After downloading, the compressed archive (.zip) must be extracted before import. The extracted directory contains the OVF descriptor and associated VMDK disk image files as illustrated below.

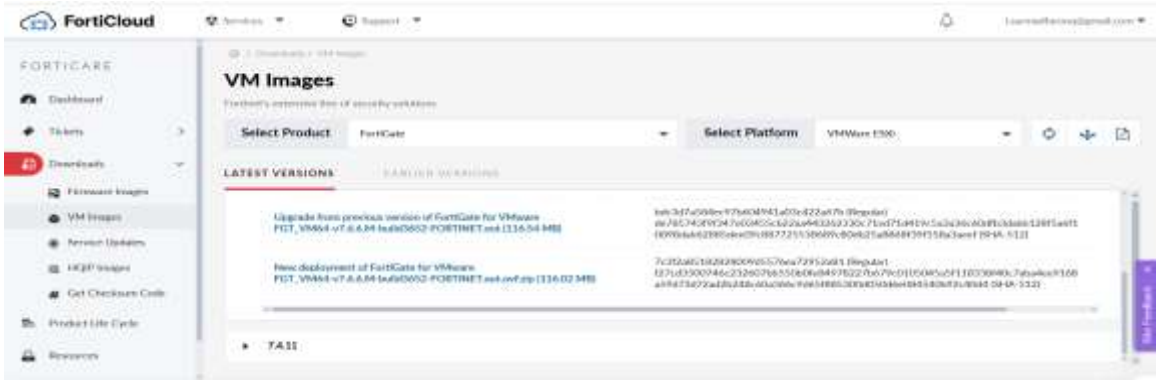


Figure 2.1 — FortiGate VM download portal on the Fortinet Support website

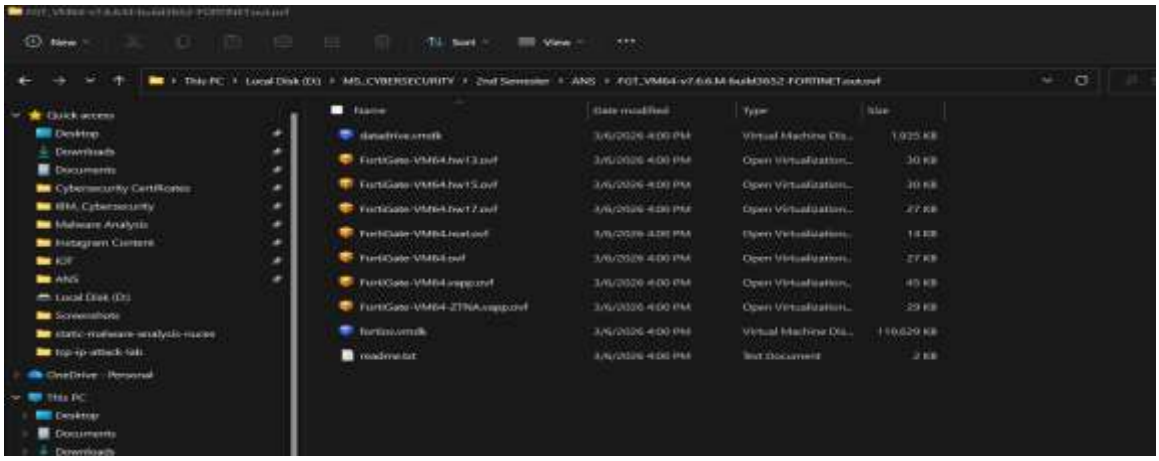


Figure 2.2 — Extracted FortiGate VM archive contents showing OVF and VMDK files

> **Key Point:**
> Always verify the downloaded archive checksum (MD5/SHA256) against values published on the Fortinet portal before deployment to ensure image integrity.

2.2 Step 2 — Configure Virtual Network Interfaces in VMware

FortiGate requires dedicated virtual network adapters for each functional zone (Management, LAN, WAN, DMZ, etc.). VMware Workstation's Virtual Network Editor is used to create and configure these host-only networks with unique IP subnets.

The Virtual Network Editor is accessible via Edit > Virtual Network Editor or by launching it directly from the Windows Start Menu. Administrator privileges are required to modify network settings.

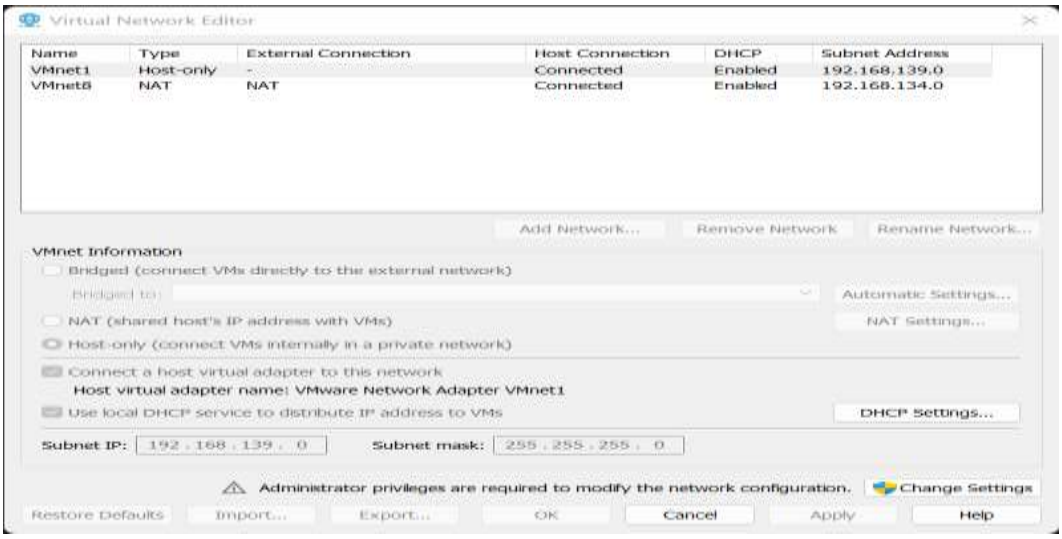


Figure 2.3 — Opening VMware Virtual Network Editor via the Edit menu

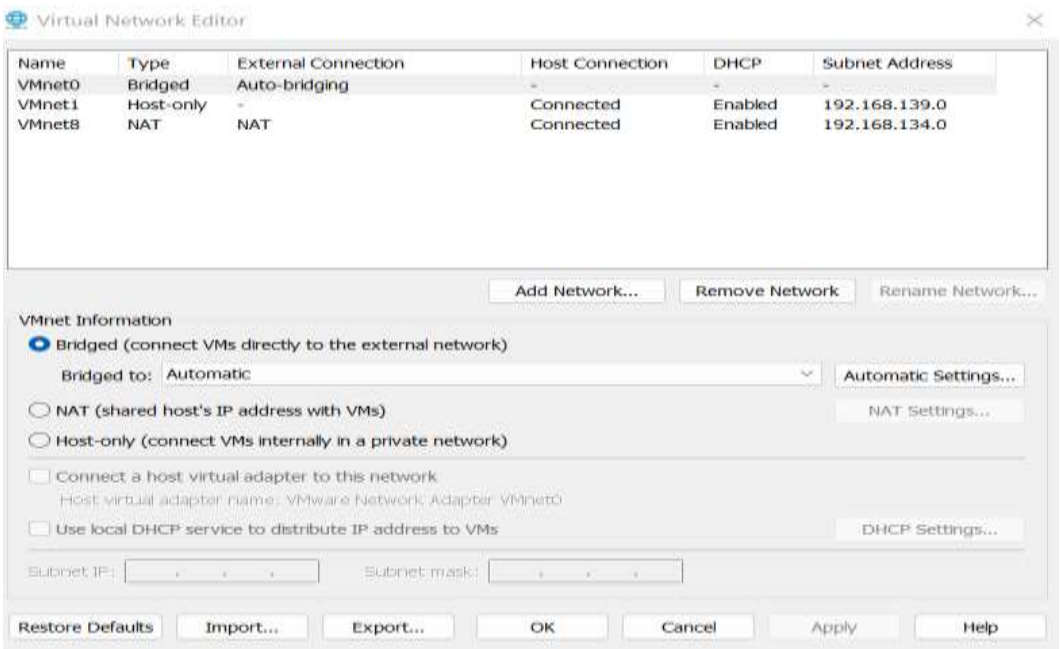


Figure 2.4 — Virtual Network Editor privilege escalation prompt requiring Administrator access

By default, VMware Workstation provides only VMNet1 (Host-Only) and VMNet8 (NAT). Additional host-only networks are created for each FortiGate interface. The following table documents the network adapter configuration used in this lab:

VMNet	Type	Subnet	DHCP	FortiGate Port
VMNet1	Host-Only	192.168.40.0/24	Disabled	port2
VMNet2	Host-Only	192.168.50.0/24	Disabled	port3
VMNet3	Host-Only	Custom Subnet	Disabled	port4
VMNet8	NAT	(VMware NAT)	Enabled	port1 (Mgmt)
VMNet11	Host-Only	Custom Subnet	Disabled	port5
VMNet12	Host-Only	Custom Subnet	Disabled	port6

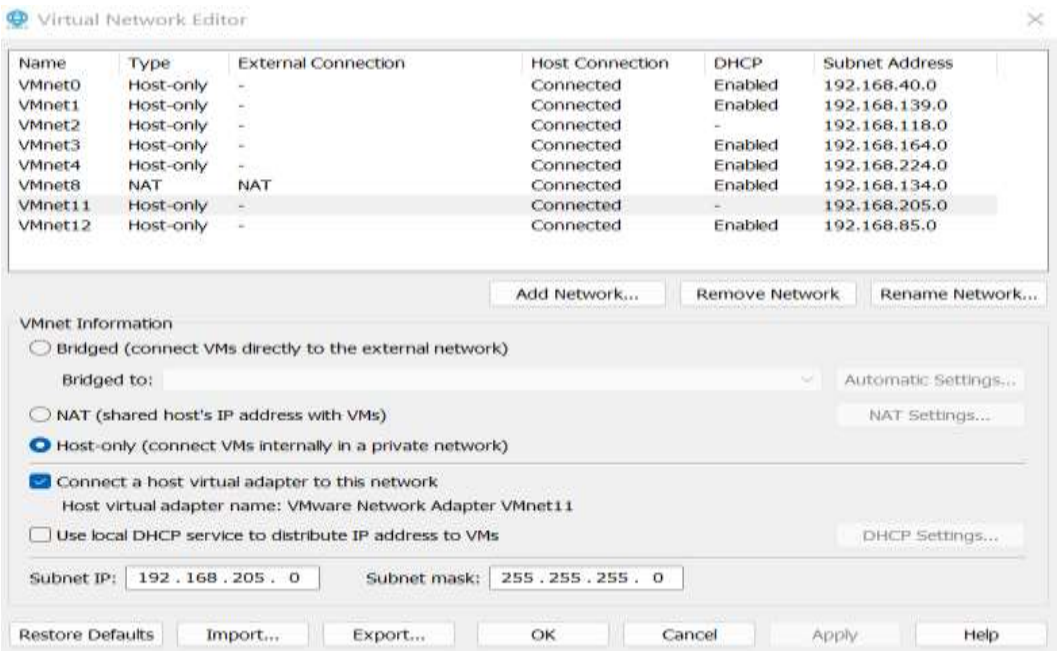


Figure 2.5 — VMware Virtual Network Editor showing Host-Only networks configured for FortiGate interfaces

2.3 Step 3 — Deploy FortiGate VM in VMware Workstation

The OVF import process provisions the virtual machine within VMware Workstation. Navigate to File > Open (Ctrl+O) or use the Home tab's "Open a Virtual Machine" option, then select the FortiGate-VM64.ovf file extracted in the previous step.

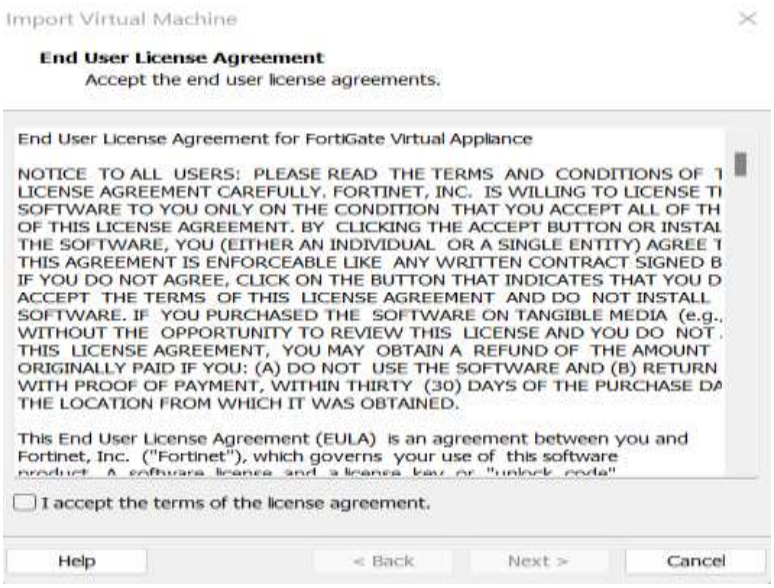


Figure 2.6 — OVF file selection dialogue during FortiGate VM import into VMware Workstation

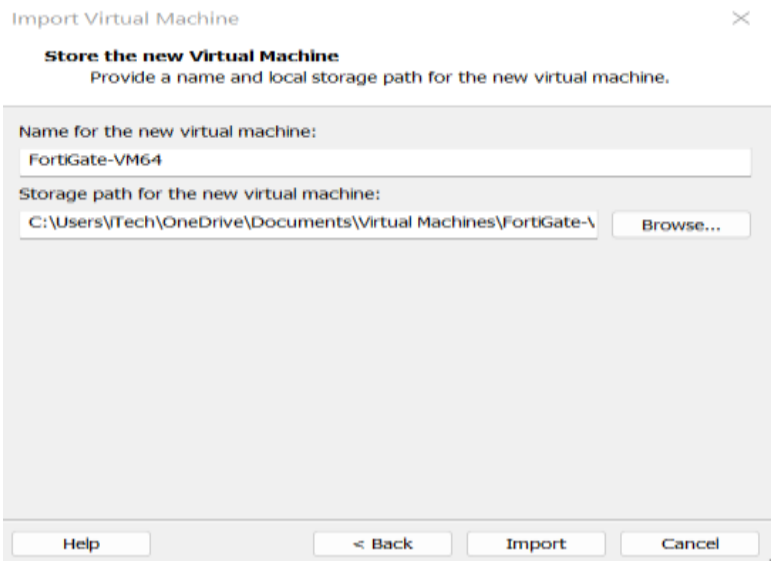


Figure 2.7 — Assigning a VM name and storage path for the imported FortiGate appliance

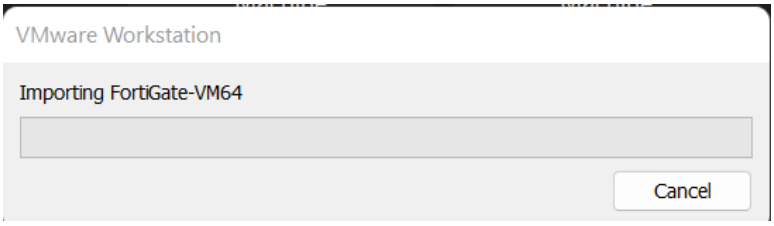


Figure 2.8 — OVF import progress indicator during VM deployment

Once import completes, the VM's hardware configuration must be edited to match the required resource allocation and network adapter assignments. Click "Edit virtual machine settings" to access the hardware configuration panel.

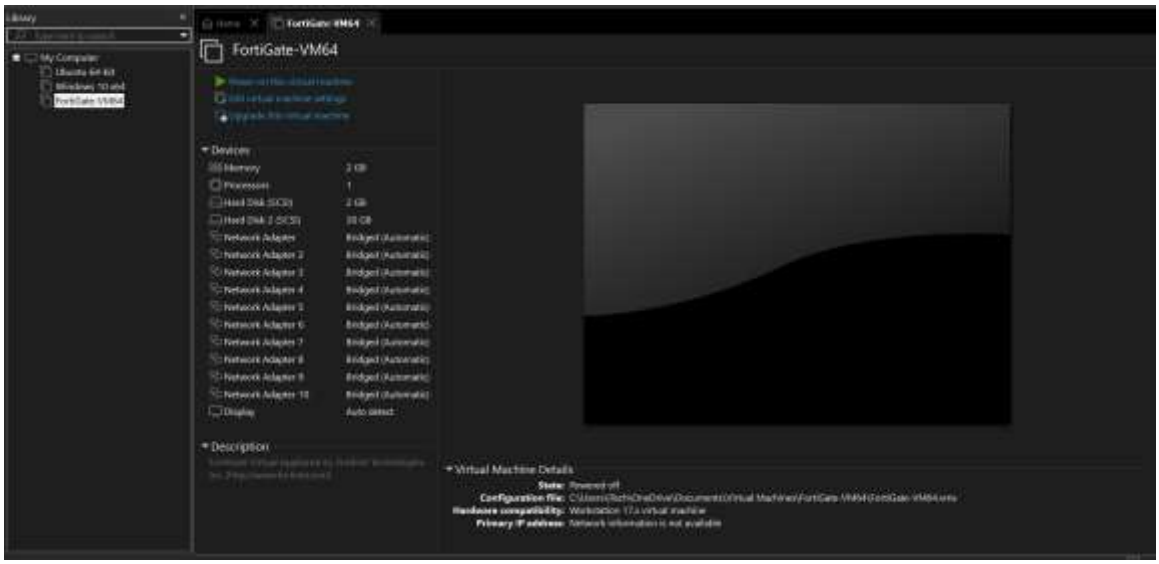


Figure 2.9 — FortiGate VM settings panel in VMware showing hardware configuration options

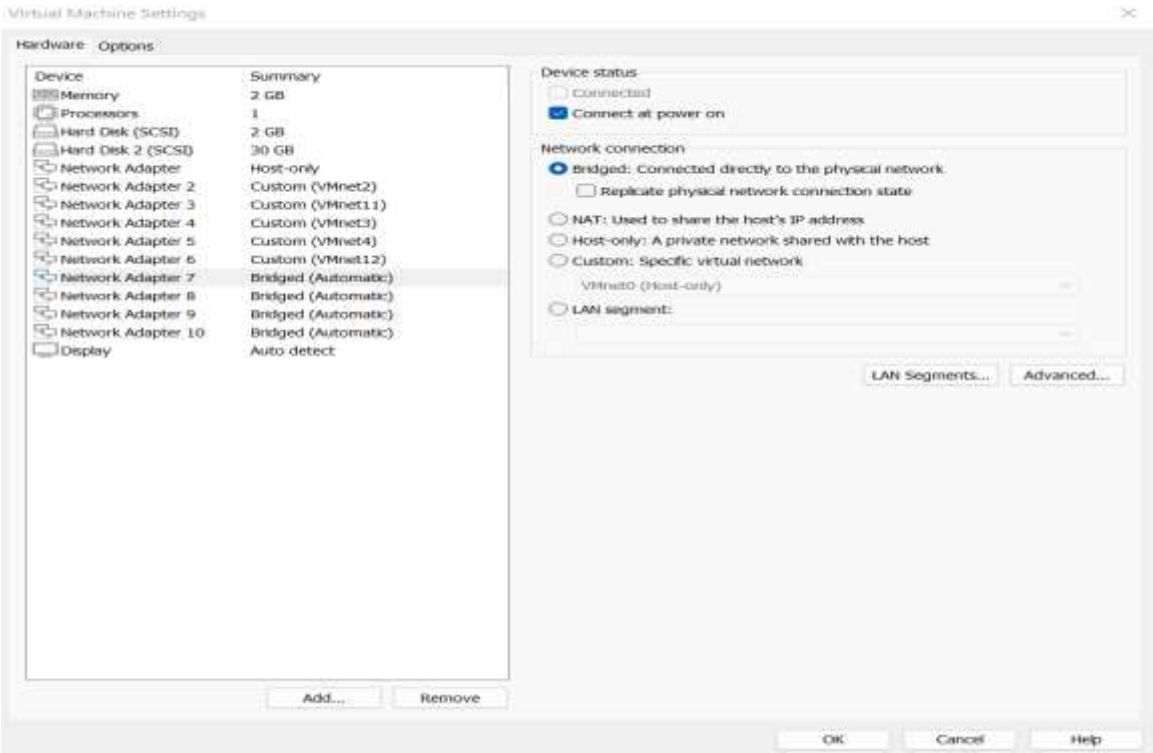


Figure 2.10 — Final VM resource allocation: 2 GB RAM, 30 GB HDD, 1 vCPU with 6 virtual NICs assigned to VMNet2–VMNet12

The six network adapters are mapped to distinct VMware virtual networks corresponding to each FortiGate logical interface (port2 through port6, plus the management port). This segregation enables zone-based policy enforcement — a fundamental principle of NGFW architecture.

2.4 Step 4 — Configure the Management Interface via CLI

After VM boot, the initial configuration is performed through the FortiGate CLI console within VMware. Default credentials are used for first login (username: admin; no password — press Enter to proceed, then set a new password as prompted).

Figure 2.11 — FortiGate CLI initial login prompt showing the password change procedure

Before assigning IP addresses, verify existing interface state with the following command:

```
get system interface
```

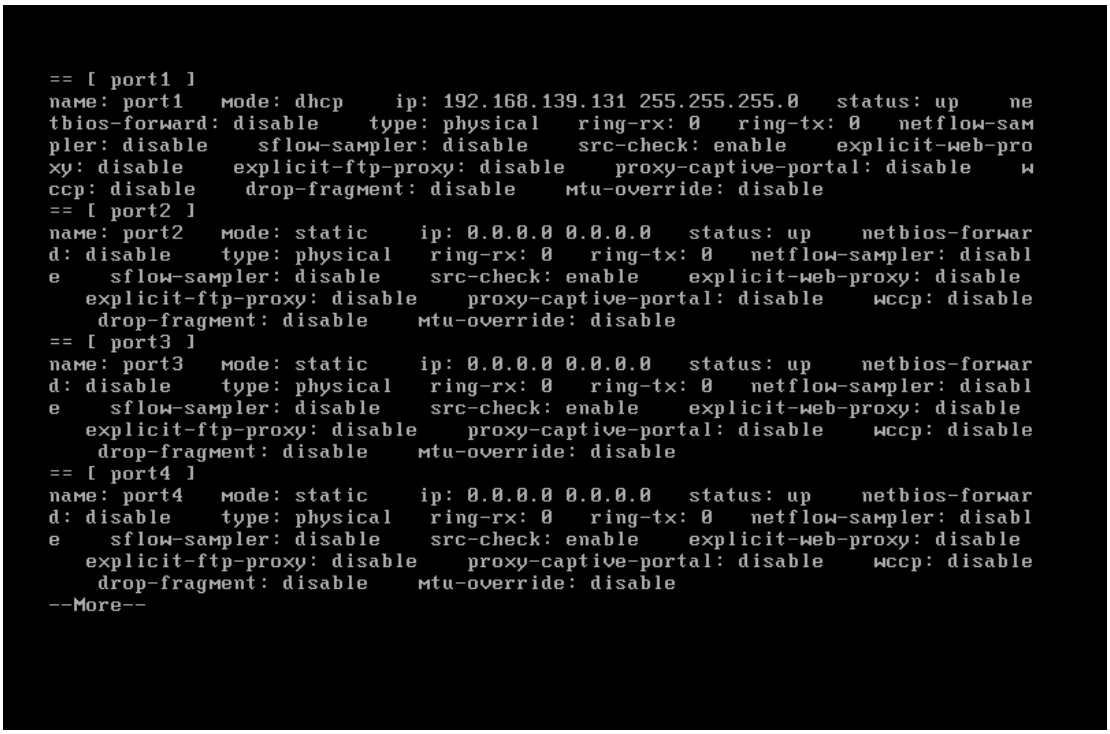


Figure 2.12 — Output of "get system interface" showing all interface names and their current states

Port1 is designated as the management interface. The following CLI command sequence assigns a static IP address and enables the required administrative access protocols:

```
config system interface
  edit port1
  set mode static
  set ip 192.168.139.131 255.255.255.0
  set allowaccess http https telnet ssh ping
end
```

```
FortiGate-VM64 #
FortiGate-VM64 # config system interface

FortiGate-VM64 (interface) # edit port1

FortiGate-VM64 (port1) # set mode static

FortiGate-VM64 (port1) # set ip 192.168.139.131 255.255.255.0

FortiGate-VM64 (port1) # set allowaccess http https telnet ssh ping

FortiGate-VM64 (port1) # end

FortiGate-VM64 #
```

Figure 2.13 — CLI commands assigning static IP 192.168.139.131/24 and enabling HTTPS/SSH on port1

Configuration can be verified with the show command:

```
show system interface
```

```
FortiGate-VM64 # show system interface
config system interface
    edit "port1"
        set vdom "root"
        set ip 192.168.139.131 255.255.255.0
        set allowaccess ping https ssh http telnet
        set type physical
        set snmp-index 1
    next
    edit "port2"
        set vdom "root"
        set type physical
        set snmp-index 2
    next
    edit "port3"
        set vdom "root"
        set type physical
        set snmp-index 3
    next
    edit "port4"
        set vdom "root"
        set type physical
        set snmp-index 4
    next
--More-- _
```

Figure 2.14 — Output of "show system interface" confirming the static IP assignment on port1

2.5 Step 5 — Access FortiGate GUI

Before opening the browser-based GUI, connectivity to the management interface is validated via ping from the host machine:

```
execute ping 192.168.139.131
```

```
FortiGate-VM64 # execute ping 192.168.139.131
PING 192.168.139.131 (192.168.139.131): 56 data bytes
64 bytes from 192.168.139.131: icmp_seq=0 ttl=255 time=0.2 ms
64 bytes from 192.168.139.131: icmp_seq=1 ttl=255 time=0.2 ms
64 bytes from 192.168.139.131: icmp_seq=2 ttl=255 time=0.2 ms
64 bytes from 192.168.139.131: icmp_seq=3 ttl=255 time=0.2 ms
64 bytes from 192.168.139.131: icmp_seq=4 ttl=255 time=0.2 ms

--- 192.168.139.131 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.2/0.2/0.2 ms

FortiGate-VM64 # _
```

Figure 2.15 — Ping test confirming reachability of FortiGate management IP 192.168.139.131

With confirmed reachability, the FortiGate web GUI is accessed over HTTPS. A self-signed certificate warning may appear in the browser — this is expected in lab environments and can be bypassed by adding a browser exception.

GUI URL: https://192.168.134.10

Username: admin

Password: [configured during CLI initialisation]

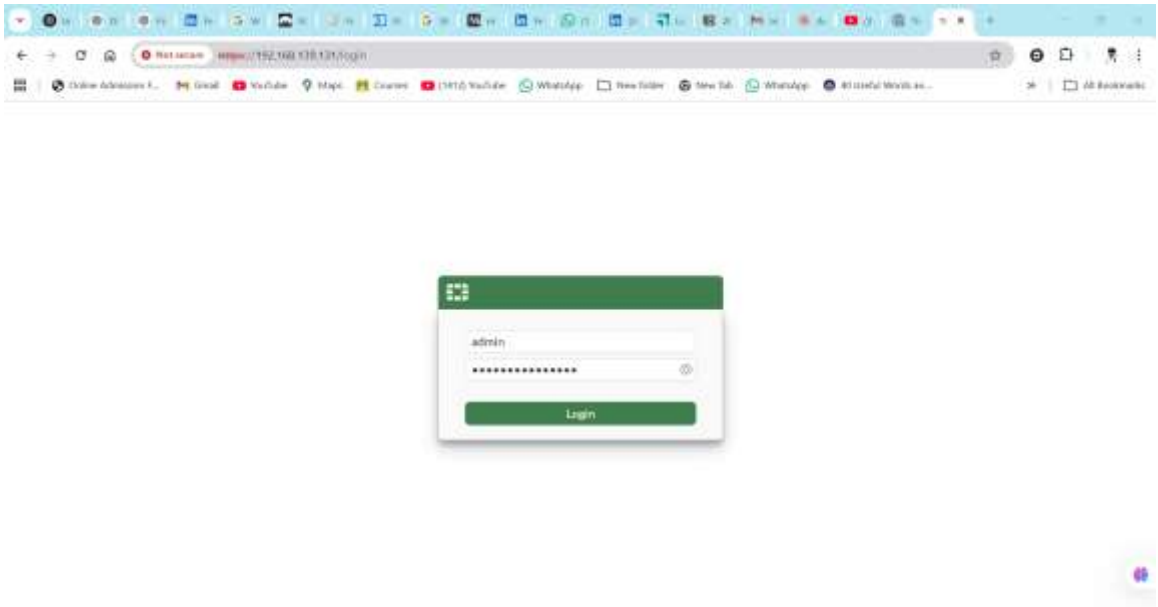


Figure 2.16 — FortiGate HTTPS login page accessed from the host browser

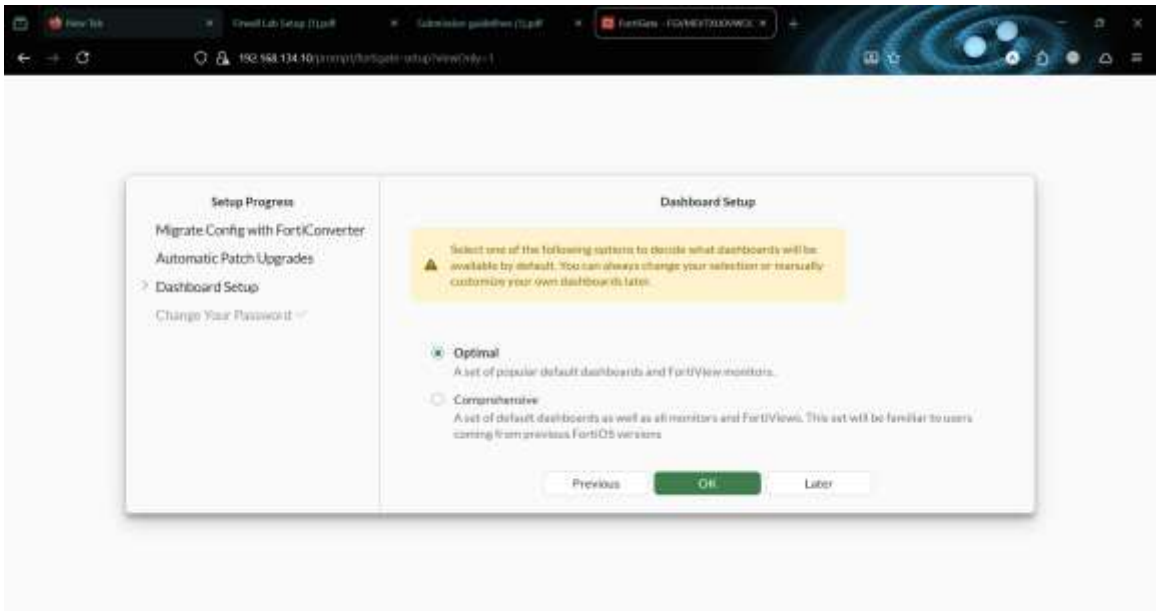


Figure 2.17 — Initial setup wizard: hostname configuration, firmware upgrade, and dashboard layout selection

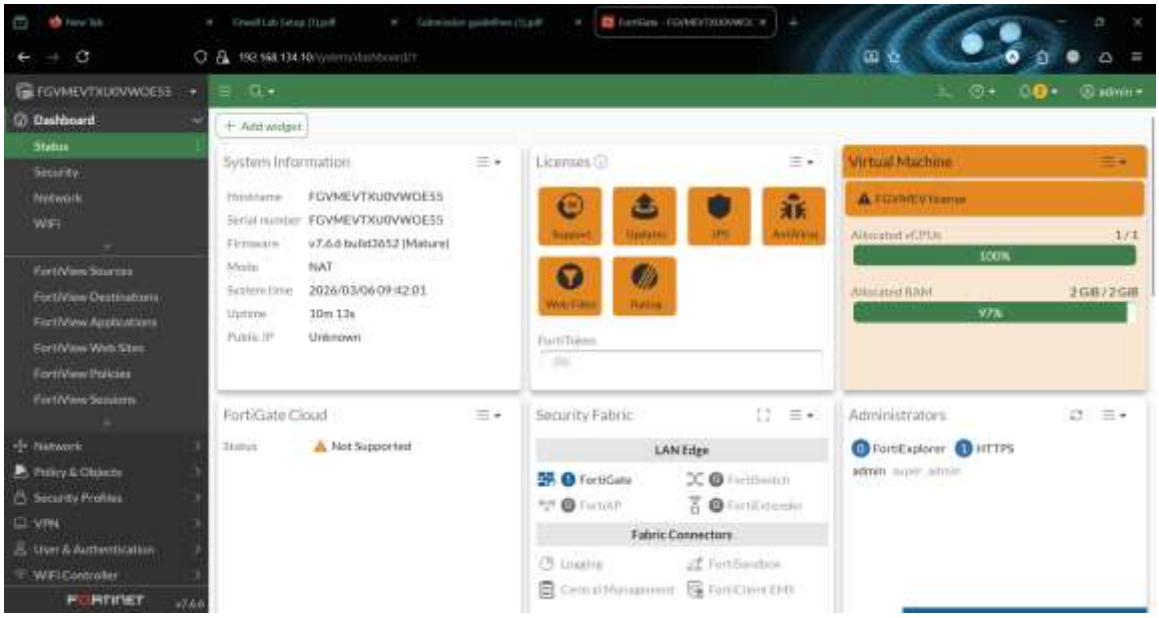


Figure 2.18 — FortiGate main dashboard after completing initial setup showing system overview

2.6 Step 6 — Configure WAN and LAN Interfaces

A properly segmented FortiGate deployment requires at minimum three logical zones: WAN (internet-facing), LAN (internal trusted network), and Management. The network topology implemented in this lab follows a standard enterprise edge firewall model:

Network Tier	Device / Component

Internet	External WAN connection from ISP
ISP Modem / CPE	Provides public IP, gateway, and DNS to FortiGate WAN port
FortiGate NGFW	Policy enforcement point — port2 (WAN), port3 (LAN), port1 (Mgmt)
Office Switch	Layer-2 distribution to endpoint devices
Employee PCs	Internal clients receiving DHCP from FortiGate

Navigate to Network > Interfaces in the GUI to configure each interface. Port2 (WAN) is set to the public IP provided by the ISP with manual addressing mode. Port3 (LAN) is assigned a private RFC1918 subnet with DHCP server enabled for client distribution.

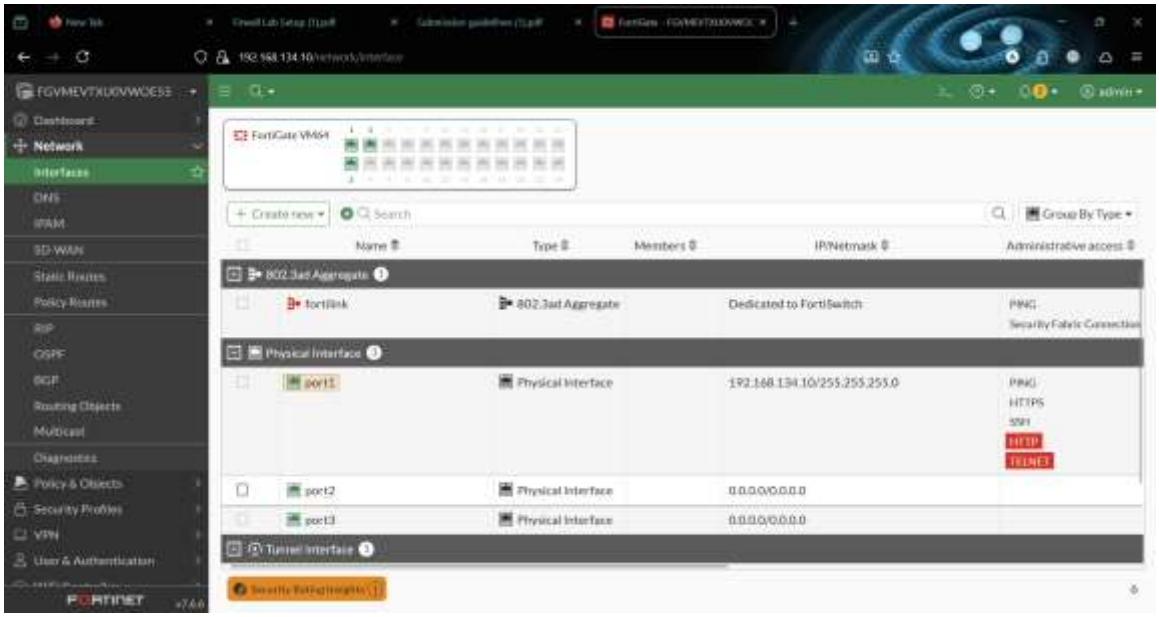


Figure 2.19 — FortiGate Network > Interfaces listing all configured ports

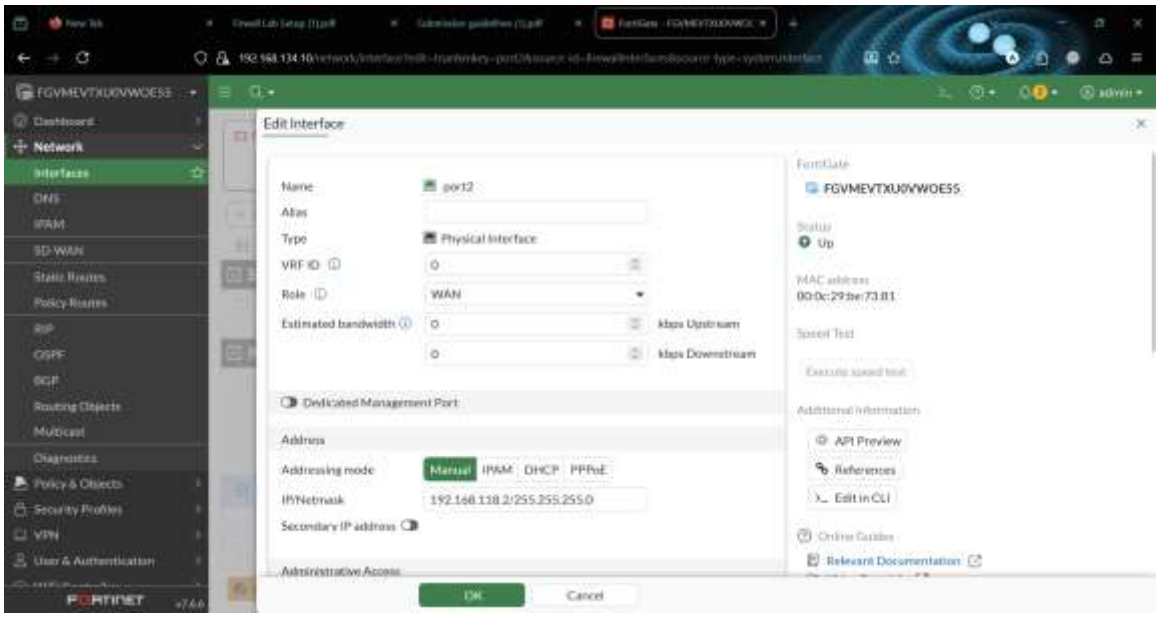


Figure 2.20 — WAN interface (port2) configuration: Manual addressing mode with ISP-provided public IP

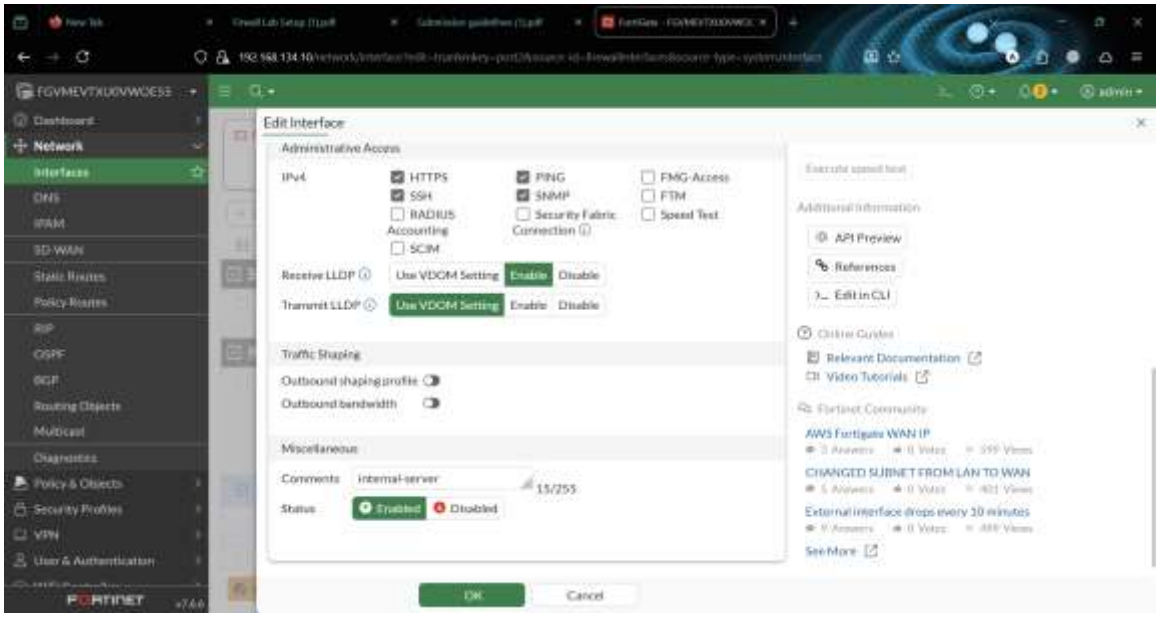


Figure 2.21 — Administrative access settings for the WAN interface

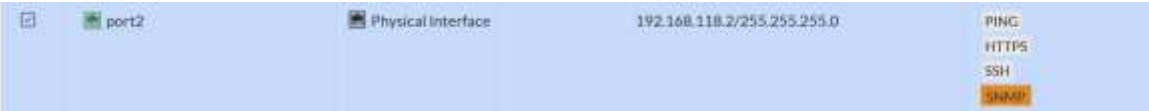


Figure 2.22 — WAN interface configuration save confirmation banner

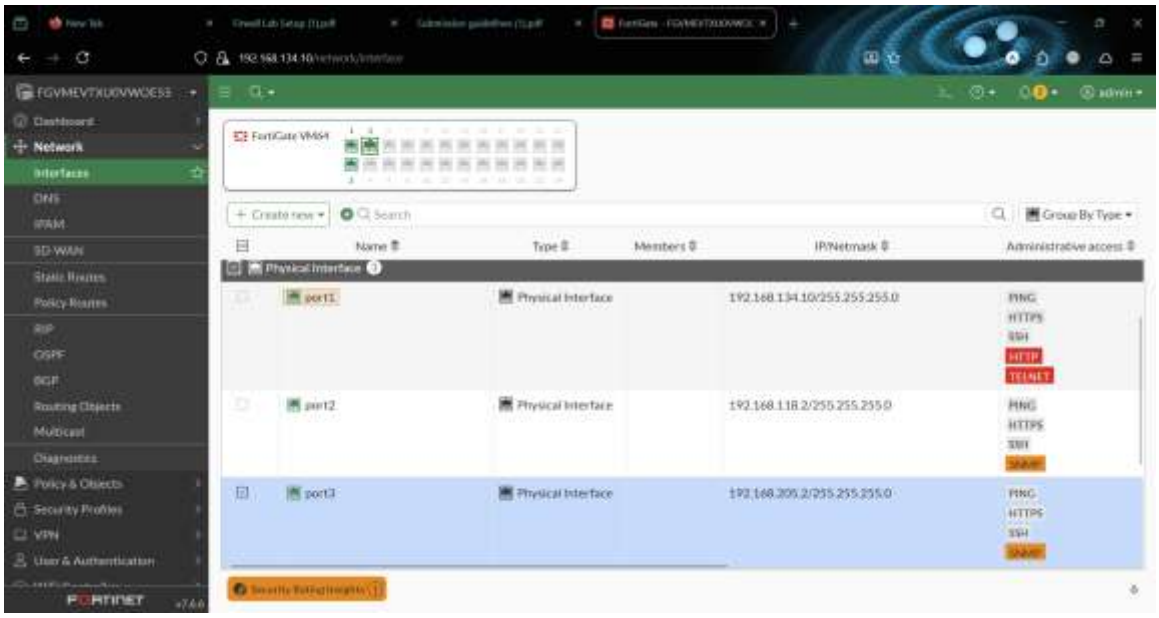


Figure 2.23 — LAN interface (port3) configuration with internal private IP and LAN role assignment

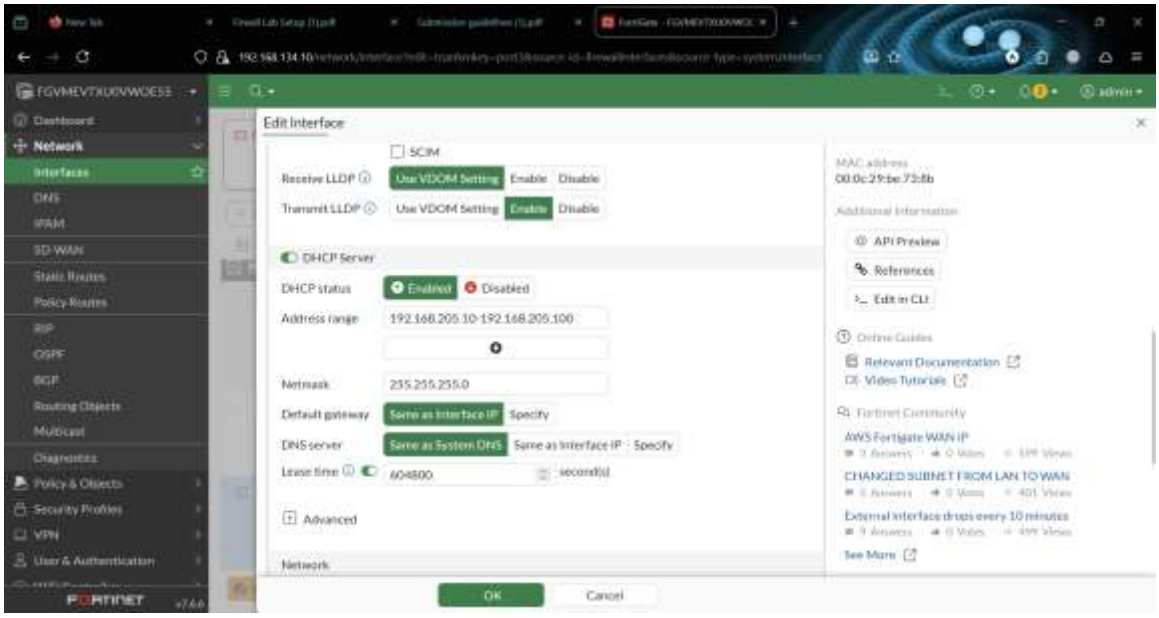


Figure 2.24 — DHCP server enabled on LAN interface for automatic IP distribution to internal clients

2.6.1 Add a Default Static Route

A default route (0.0.0.0/0) is required to direct all internet-bound traffic to the ISP gateway. Navigate to Network > Static Routes and create a new route with the following parameters:

Parameter	Value
Destination	Subnet — 0.0.0.0 / 0.0.0.0 (default route)
Gateway IP	ISP-provided gateway IP address

Interface	port2 (WAN / internet-facing interface)
Administrative Distance	10 (default for static routes)

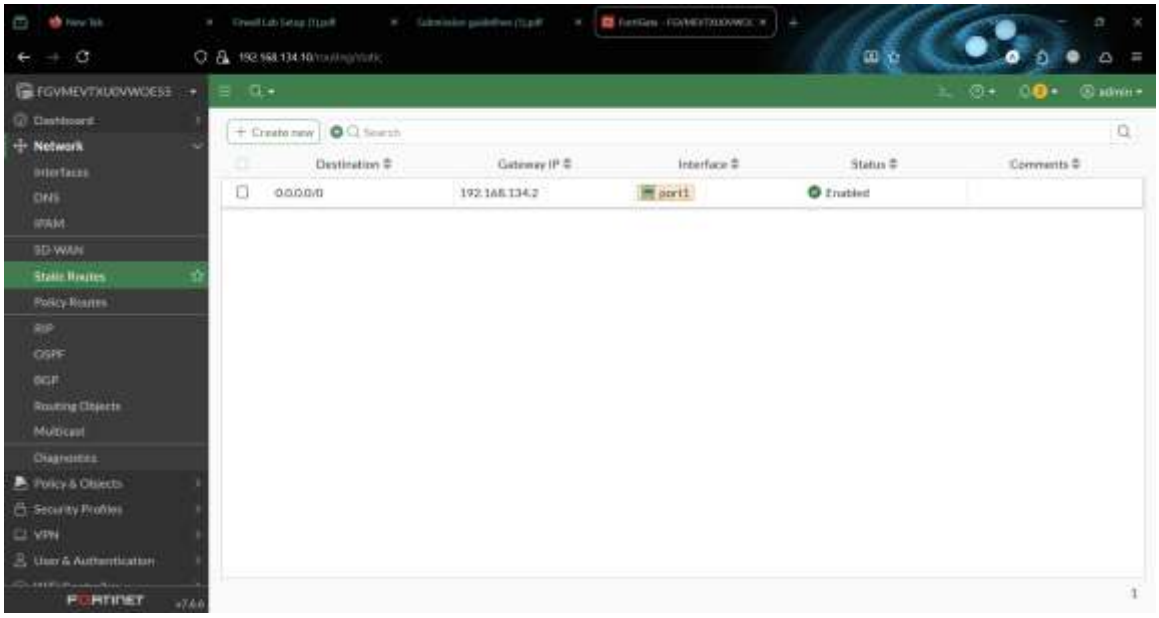


Figure 2.25 — FortiGate Static Routes page — creating a new default route

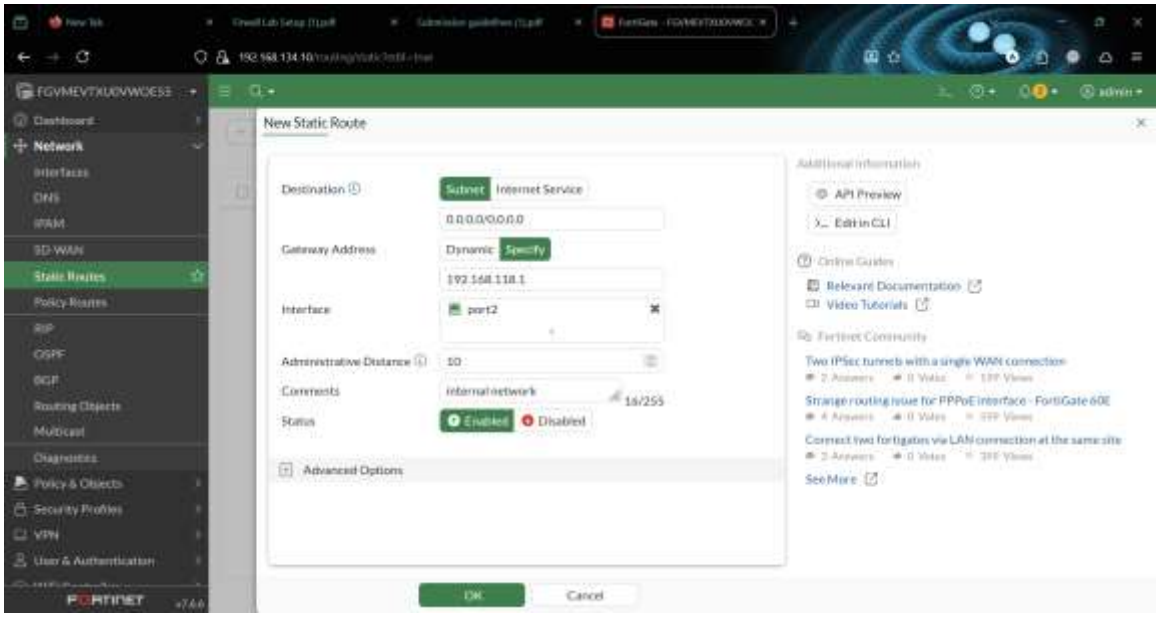


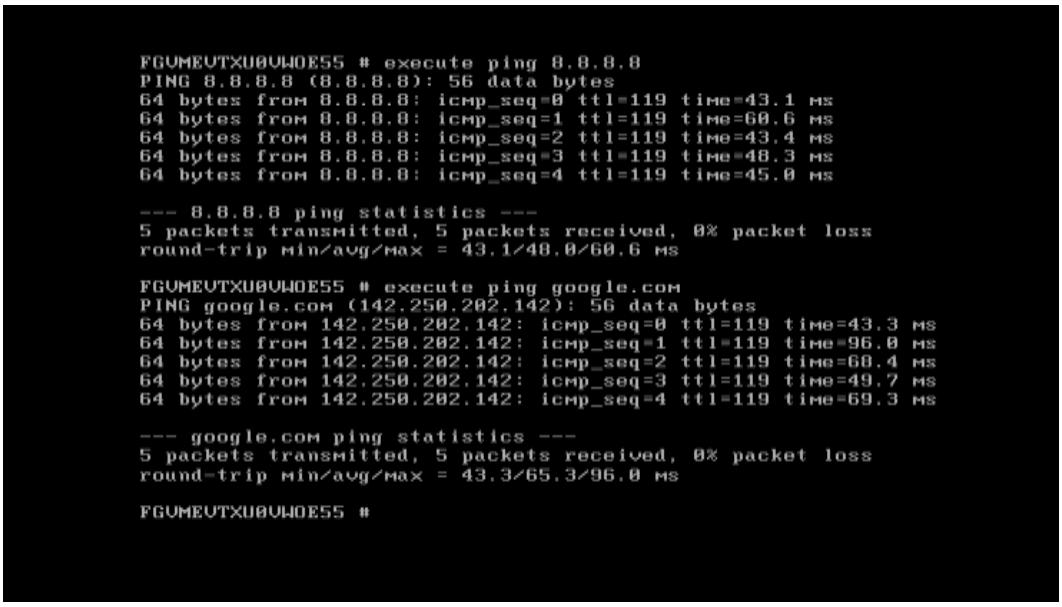
Figure 2.26 — Default route configuration: destination 0.0.0.0/0 via ISP gateway on port2

Destination	Gateway IP	Interface	Status	Comments
0.0.0.0/0	192.168.134.2	port1	Enabled	
0.0.0.0/0	192.168.118.1	port2	Enabled	internal network

Figure 2.27 — Static route saved successfully — default route now active in routing table

2.7 Step 7 — Connectivity Test

Final connectivity validation confirms that clients on the LAN segment can reach internet destinations through the FortiGate. A ping test from the LAN client to an external IP verifies end-to-end path functionality through the firewall.



```
FGVMEVTXU8VWOE55 # execute ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=119 time=43.1 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=119 time=60.6 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=119 time=43.4 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=119 time=48.3 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=119 time=45.0 ms

--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 43.1/48.0/60.6 ms

FGVMEVTXU8VWOE55 # execute ping google.com
PING google.com (142.250.202.142): 56 data bytes
64 bytes from 142.250.202.142: icmp_seq=0 ttl=119 time=43.3 ms
64 bytes from 142.250.202.142: icmp_seq=1 ttl=119 time=96.0 ms
64 bytes from 142.250.202.142: icmp_seq=2 ttl=119 time=68.4 ms
64 bytes from 142.250.202.142: icmp_seq=3 ttl=119 time=49.7 ms
64 bytes from 142.250.202.142: icmp_seq=4 ttl=119 time=69.3 ms

--- google.com ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 43.3/65.3/96.0 ms

FGVMEVTXU8VWOE55 #
```

Figure 2.28 — Successful ping test from LAN client through FortiGate to internet, confirming WAN/LAN routing is operational

3. Task 2 — GUI Access Configuration

Securing the management plane is as critical as securing the data plane. This task hardens the FortiGate administrative interface by verifying HTTPS-only access, configuring strong admin credentials, and confirming authenticated GUI login.

3.1 Step 1 — Enable and Verify HTTPS GUI Access

HTTPS access was pre-enabled during the CLI management interface configuration (set allowaccess https) in Task 1. This step verifies the GUI setting through the web interface and ensures no insecure protocols (HTTP, Telnet) are unnecessarily exposed on the management interface.

Navigate to Network > Interfaces > port1 > Edit and examine the Administrative Access section.

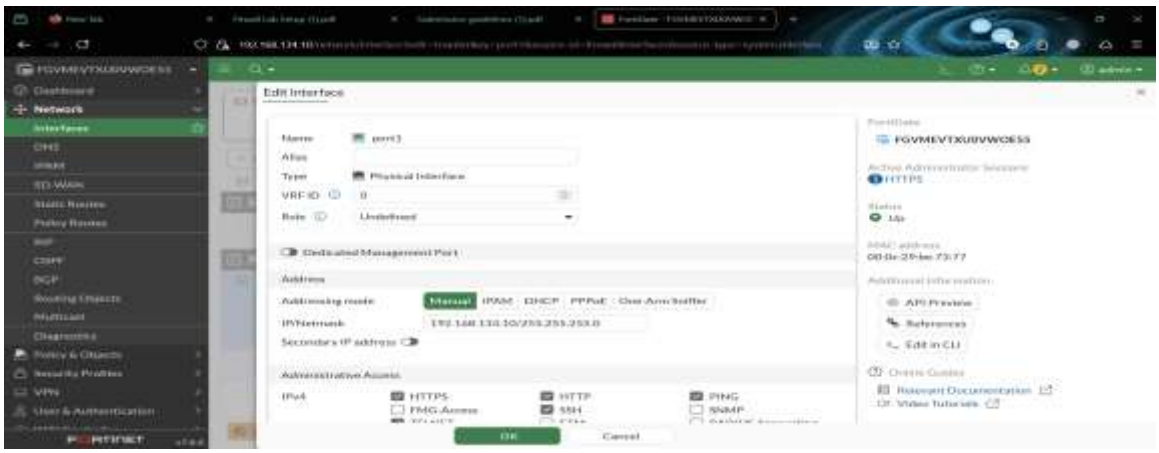


Figure 3.1 — Network > Interfaces panel showing port1 selected for editing

Figure 3.2 — Administrative Access settings confirming HTTPS checkbox is enabled on port1

Figure 3.3 — Saving interface configuration after verifying HTTPS access setting

> **Security Best Practice:**
> In production environments, disable HTTP, Telnet, and SNMP on management interfaces. Restrict administrative access to specific trusted source IP addresses using Trusted Hosts in the administrator profile.

3.2 Step 2 — Configure Admin Credentials

Default administrator credentials represent a critical attack surface. FortiGate ships with a default "admin" account. This step reconfigures the admin profile with a strong password and documents the process.

Navigate to System > Administrators to view the accounts panel:

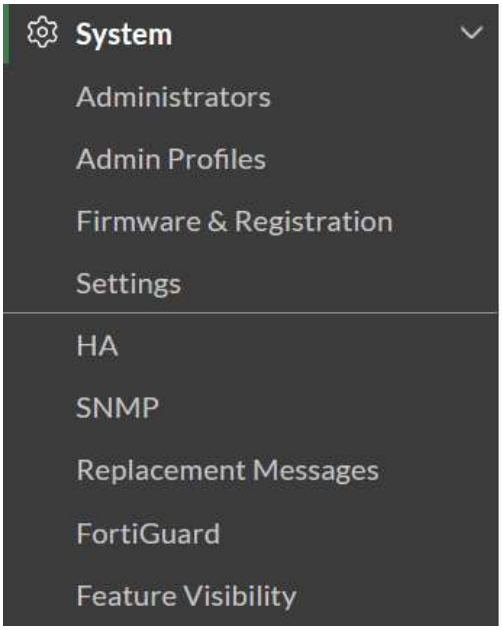


Figure 3.4 — System > Administrators navigation in the FortiGate menu

Figure 3.5 — Administrators list showing the default admin account

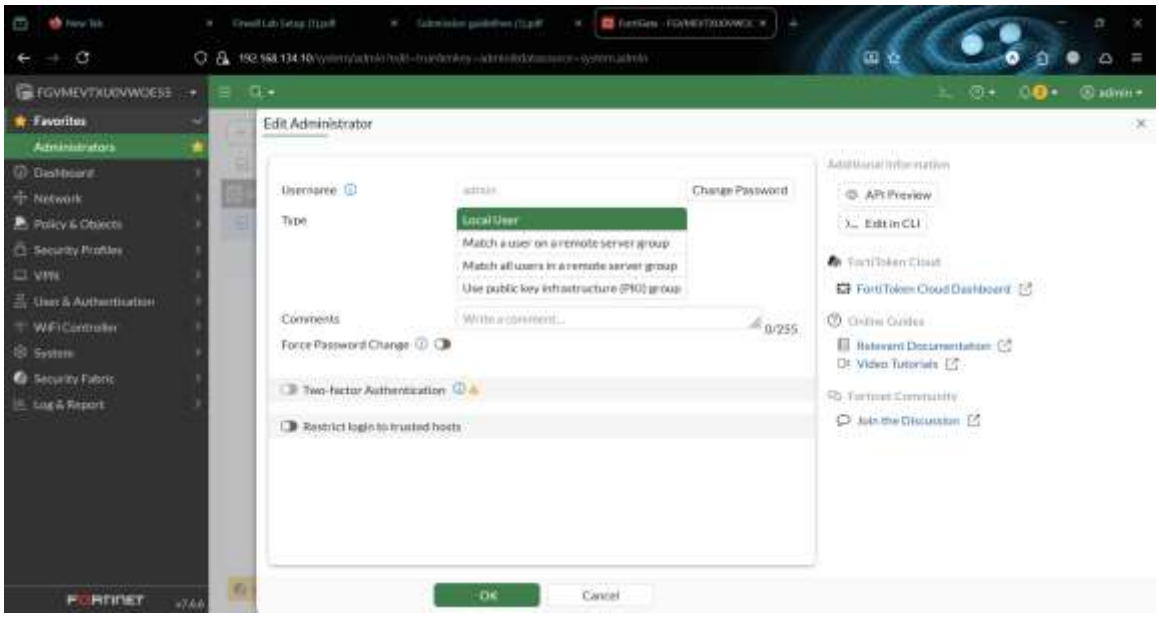


Figure 3.6 — Admin account edit panel: updating password and optionally adding Trusted Host restrictions

Setting	Recommended Configuration
Username	admin (can be renamed for obscurity)
Password	Minimum 12 characters; mix upper/lower/digit/special characters
Admin Profile	super_admin for primary account; restrict with custom profiles for secondary accounts
Trusted Hosts	Restrict to management workstation IP or subnet (e.g., 192.168.139.100/32)
Two-Factor Auth	Enable FortiToken or email-based 2FA in production environments

3.3 Step 3 — Verify Login

The login verification step confirms that updated credentials function correctly and that the GUI is accessible only via HTTPS. Both the login page and successful authenticated dashboard are captured as evidence.

Figure 3.7 — FortiGate HTTPS login page with updated admin credentials being entered

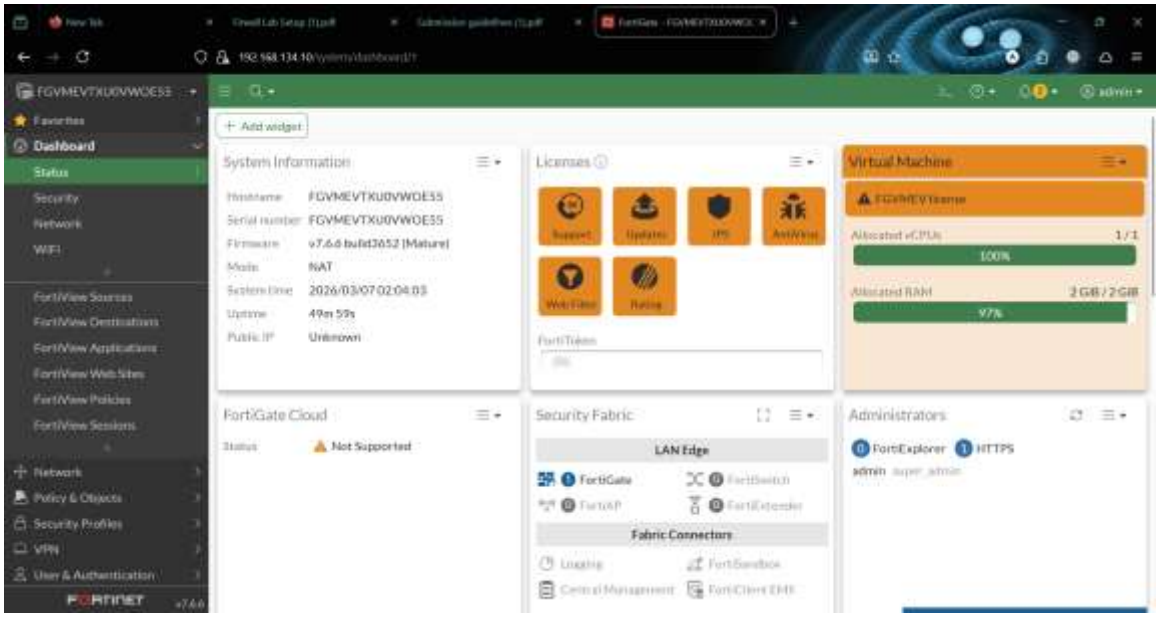


Figure 3.8 — Successful authenticated session — FortiGate dashboard accessible with new admin credentials

4. Task 3 — Policy Implementation

This task implements three distinct security policies on FortiGate that together form a layered defence-in-depth architecture: an IPv4 firewall policy for baseline traffic control, an IPv4 DoS anomaly protection policy for volumetric attack mitigation, and a web content filtering policy for application-layer access control.

4.1 Policy A — IPv4 Firewall Policy (Internet Access)

Firewall policies are the core mechanism by which FortiGate controls traffic flows between network segments. Each policy is a rule consisting of match criteria (source, destination, service) and an action (accept, deny, IPSec). A NAT-enabled outbound policy is required to allow internal LAN clients to reach the internet.

Navigate to Policy & Objects > Firewall Policy and create a new policy:

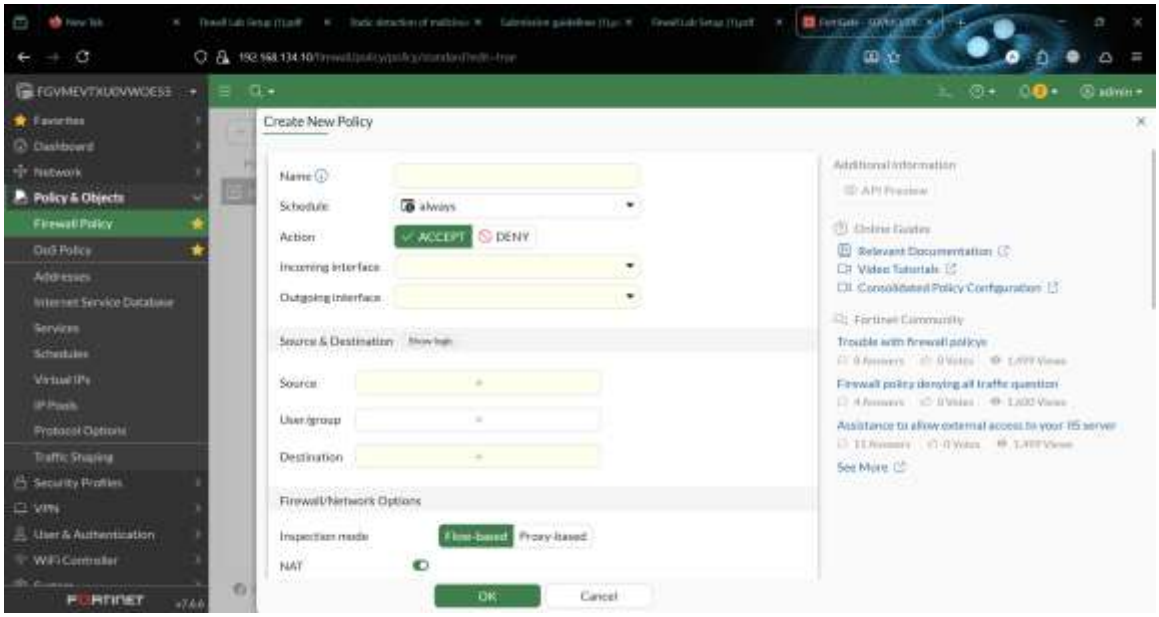


Figure 4.1 — Policy & Objects > Firewall Policy page showing the policy creation interface

Policy Parameter	Configured Value
Policy Name	Internet-Traffic
Incoming Interface	port3 (LAN / Internal network)
Outgoing Interface	port2 (WAN / Internet-facing)

Source	all (or specify internal subnet for tighter control)
Destination	all
Service	ALL (HTTP, HTTPS, DNS, etc.)
Action	ACCEPT
NAT	Enabled — Use Outgoing Interface Address (PAT/masquerade)
Security Profiles	None applied at baseline (added in subsequent policies)

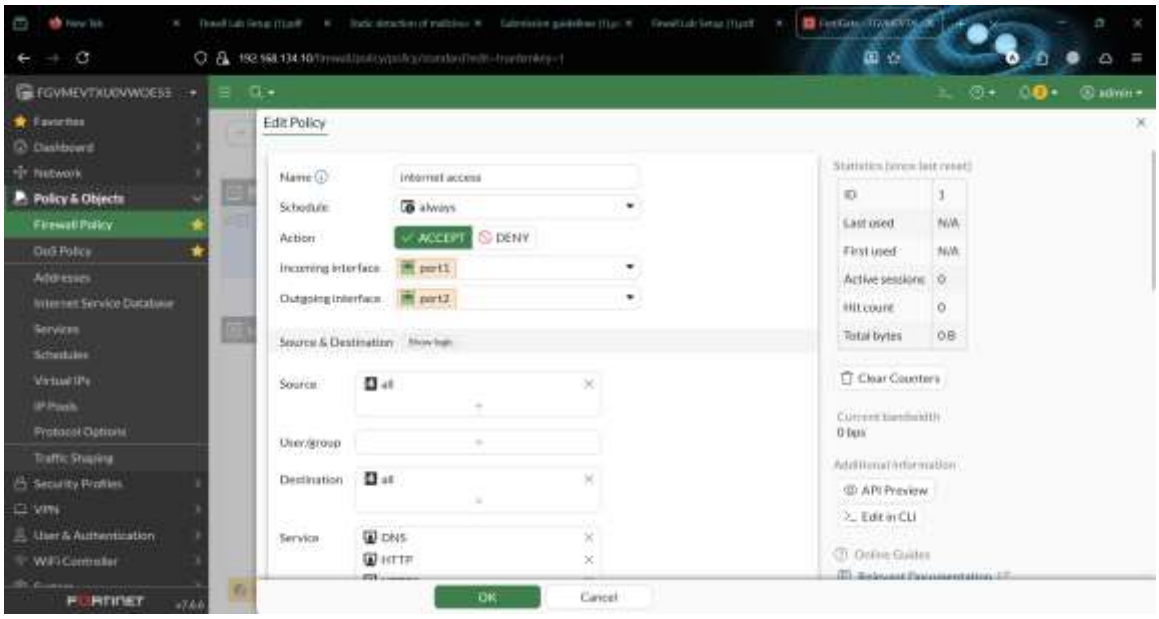


Figure 4.2 — IPv4 firewall policy configuration: name, interfaces, source, and destination settings

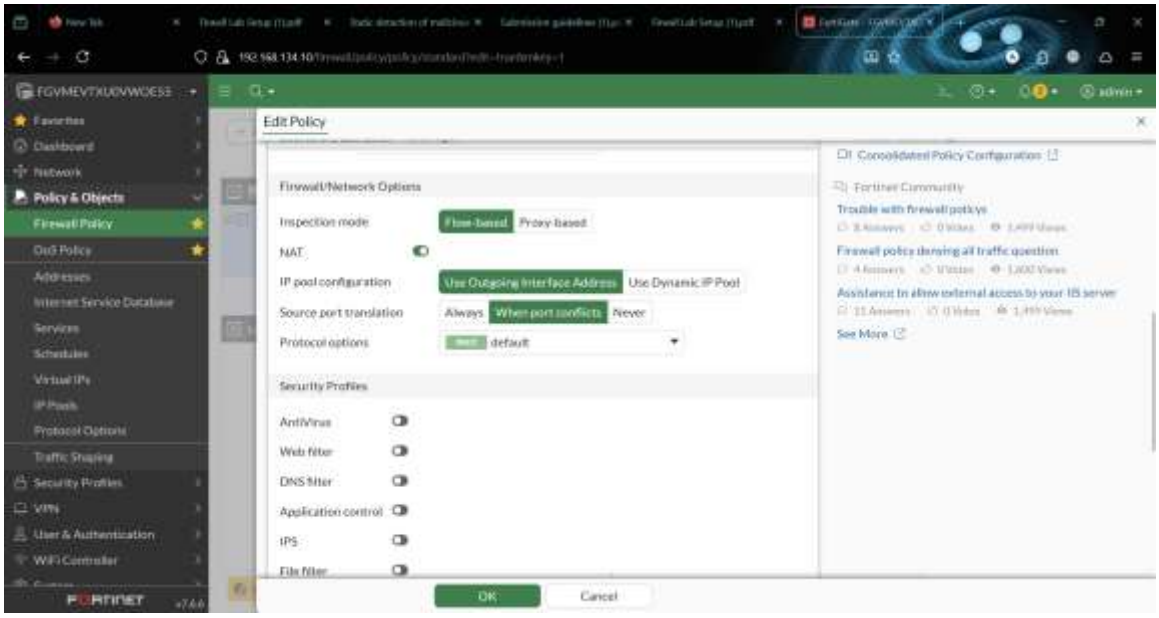


Figure 4.3 — Policy action set to ACCEPT with NAT enabled using outgoing interface address

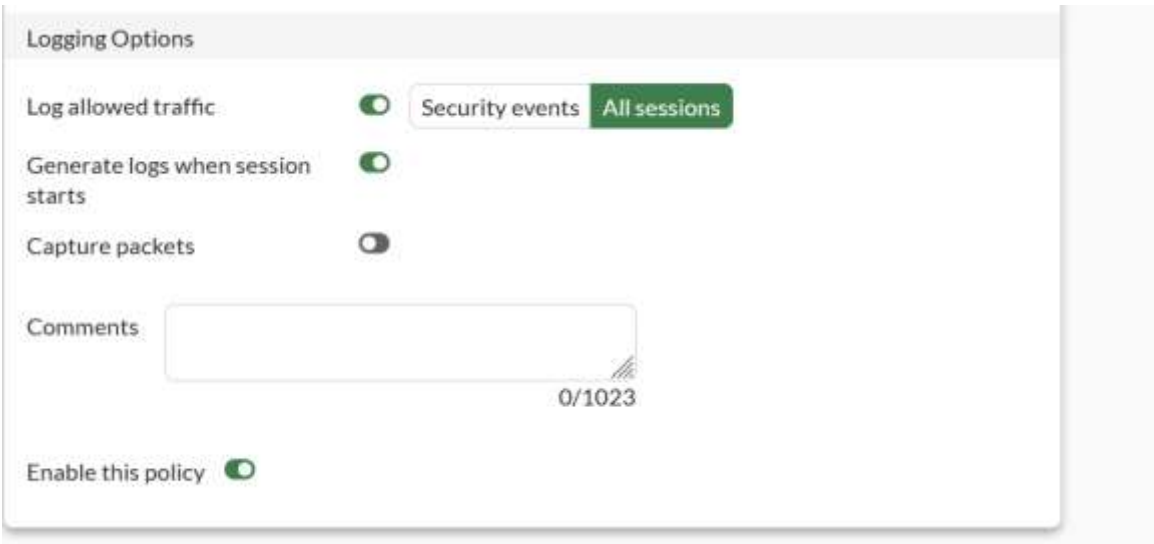


Figure 4.4 — Additional policy options and service group configuration

Figure 4.5 — "Internet-Traffic" policy saved and active in the firewall policy table

Once saved, the policy appears in the Policy & Objects > Firewall Policy table. The green indicator confirms the policy is enabled and actively evaluated against matching traffic flows.

4.2 Policy B — IPv4 DoS Anomaly Protection Policy

Denial-of-Service (DoS) attacks represent a persistent threat to network availability. FortiGate's DoS policy module provides stateful anomaly detection at Layers 3 and 4, applying threshold-based controls to detect and block volumetric attack patterns including SYN floods, UDP floods, ICMP sweeps, and port scans.

Navigate to Policy & Objects > IPv4 DoS Policy and create a new policy:

4.2.1 Layer 3 (IP-Level) Anomaly Profiles

L3 anomaly profiles operate on the IP header fields and track per-source and per-destination packet rates:

Anomaly Profile	Status	Logging	Action	Description
ip_src_session	Enabled	Enabled	Block	Max concurrent sessions per source IP

ip_dst_session	Enabled	Enabled	Block	Max concurrent sessions per destination IP

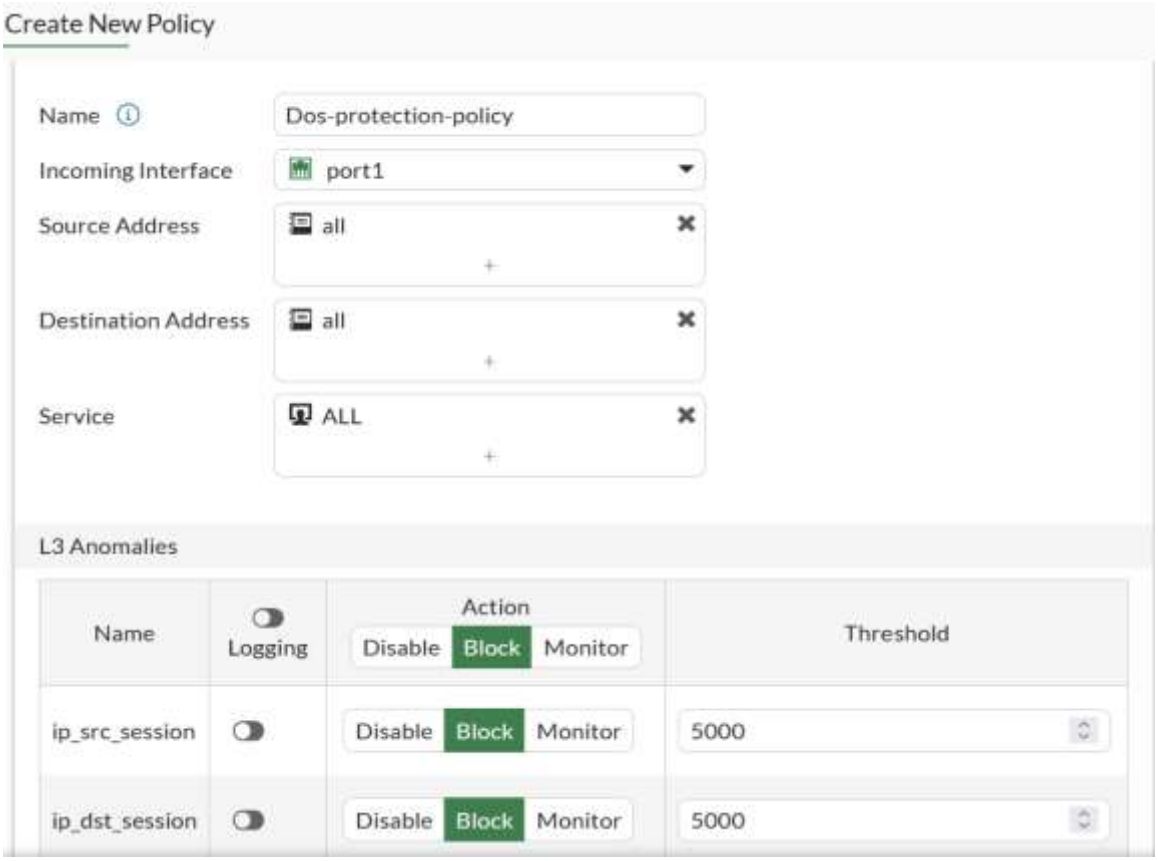


Figure 4.6 — IPv4 DoS Policy creation page showing L3 anomaly profile configuration (ip_src_session, ip_dst_session)

Figure 4.7 — Threshold and action settings for L3 IP-level anomaly profiles

4.2.2 Layer 4 (Transport-Level) Anomaly Profiles

L4 anomaly profiles inspect TCP, UDP, ICMP, and SCTP header fields to detect transport-layer attack patterns:

Anomaly Profile	Status	Log	Action	Description / Attack Vector
tcp_syn_flood	ON	ON	Block	SYN packet flood — exhausts TCP state table
tcp_port_scan	ON	ON	Block	Rapid TCP port probing for service enumeration
tcp_src_session	ON	ON	Block	Excessive TCP sessions from single source
tcp_dst_session	ON	ON	Block	Excessive TCP sessions to single destination
udp_flood	ON	ON	Block	High-rate UDP packet flood (DNS amplification)
udp_scan	ON	ON	Block	UDP port scan for service discovery
udp_src_session	ON	ON	Block	Excessive UDP sessions from single source
udp_dst_session	ON	ON	Block	Excessive UDP sessions to single destination
icmp_flood	ON	ON	Block	ICMP echo flood (Ping of Death / Smurf attack)
icmp_sweep	ON	ON	Block	ICMP sweep for network host discovery
icmp_src_session	ON	ON	Block	Excessive ICMP sessions from single source
sctp_flood	ON	ON	Block	SCTP association flood targeting telecom stacks
sctp_scan	ON	ON	Block	SCTP port scan
sctp_src_session	ON	ON	Block	Excessive SCTP associations from single source
sctp_dst_session	ON	ON	Block	Excessive SCTP associations to single destination

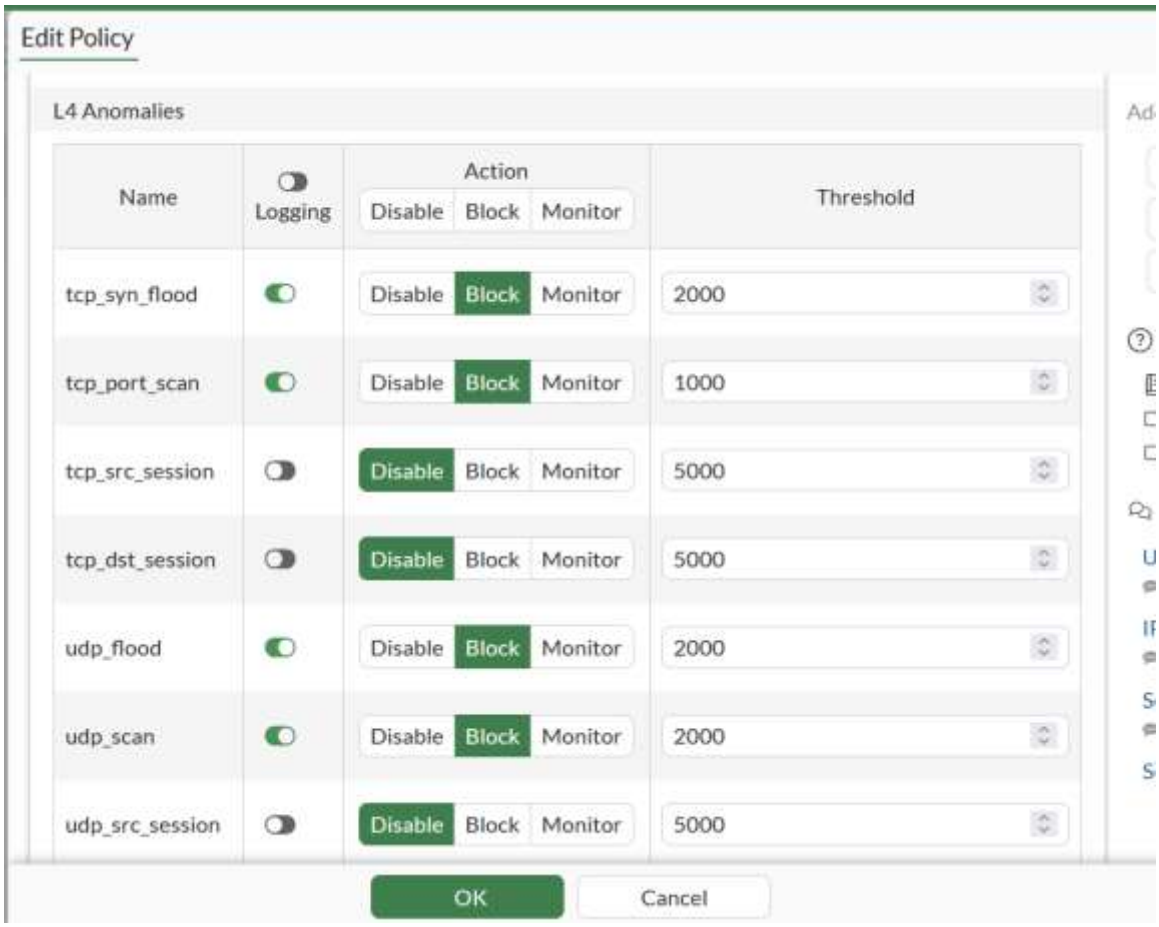


Figure 4.8 — DoS policy L4 anomaly profiles (TCP/UDP group) showing threshold and action configuration

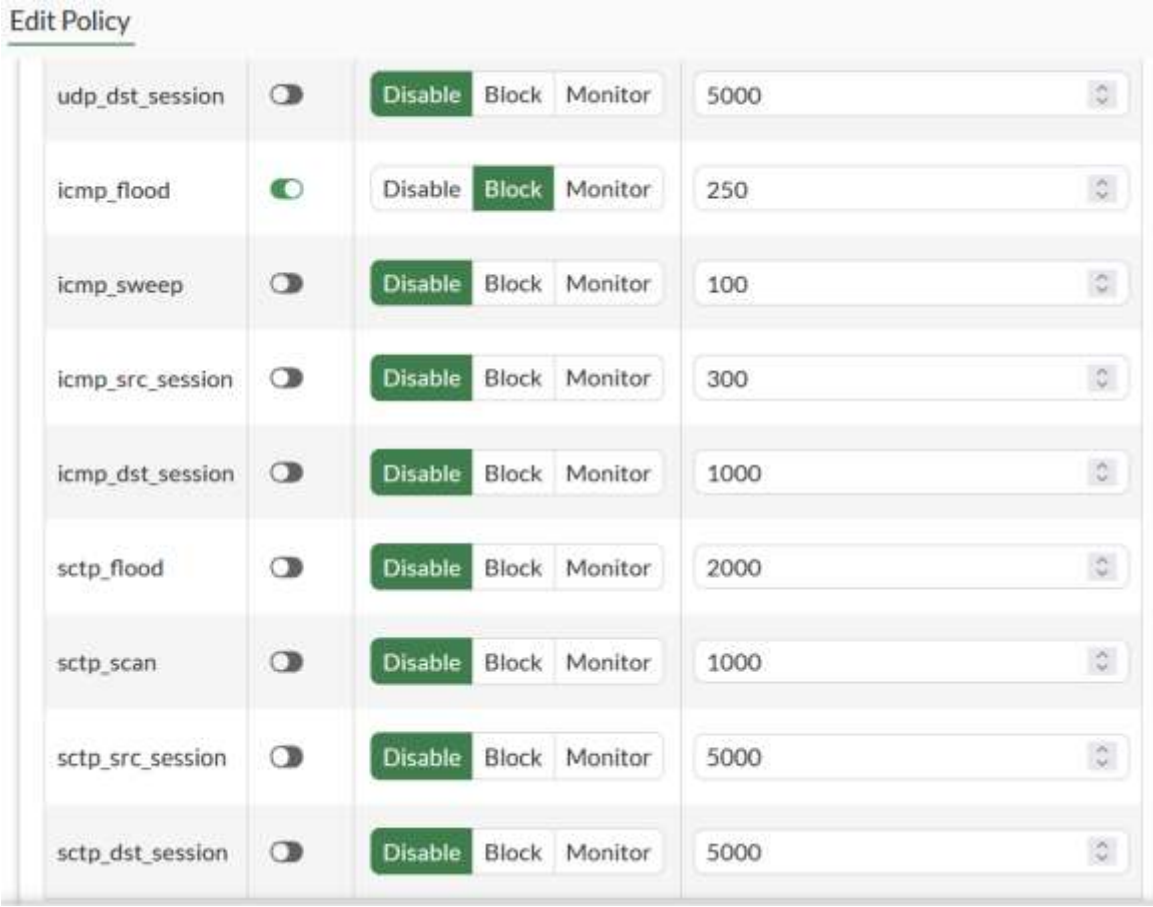


Figure 4.9 — DoS policy L4 anomaly profiles (ICMP/SCTP group) with block actions enabled

Each anomaly profile exposes four configurable parameters:

Parameter	Description
Status	Enable or Disable the anomaly profile from active evaluation
Logging	Enable to generate log entries when the profile threshold is reached or exceeded
Action	Pass — allow traffic even if threshold is reached; Block — drop packets that exceed threshold
Threshold	Number of anomalous packets or sessions per second that must be detected before the action is triggered

Figure 4.10 — DoS-Protection-Policy successfully saved and enabled in the policy table

4.3 Policy C — Web Content Filtering (URL Blocking)

Web filtering at Layer 7 (application layer) allows FortiGate to inspect HTTP/HTTPS URLs and apply category-based or explicit URL-based access controls. This capability is essential for enforcing acceptable use policies (AUP) and blocking unproductive or harmful web content in enterprise environments.

4.3.1 Enable Web Filter Feature

The Web Filter feature must first be activated in FortiGate's Feature Visibility before it becomes available in the security profile menu.

Navigate to System > Feature Visibility and enable the Web Filter toggle:

Figure 4.11 — System > Feature Visibility panel with Web Filter feature being enabled

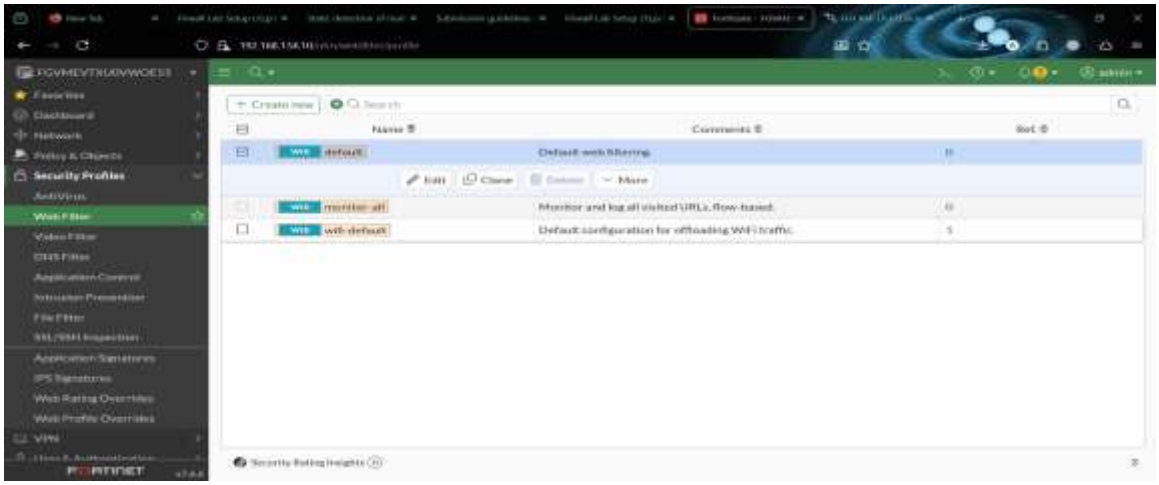


Figure 4.12 — Default Web Filter Profile activated and visible in the security profiles list

4.3.2 Configure URL Filter — Block Facebook

Navigate to Security Profiles > Web Filter and select or create a web filter profile. Within the profile, navigate to Static URL Filter > URL Filter and create a new URL entry:

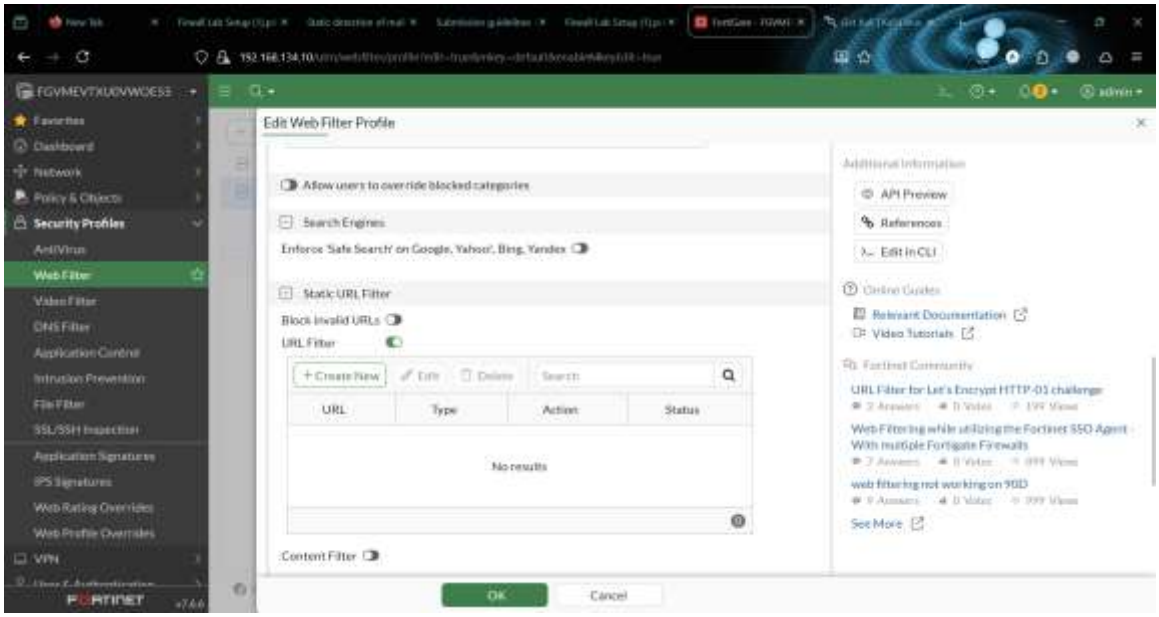


Figure 4.13 — Web Filter profile showing Static URL Filter section and the Create URL entry button

URL Filter Parameter	Configured Value
URL	facebook.com
Type	Wildcard (matches *.facebook.com and facebook.com)
Action	Block — access denied with block page shown to user
Status	Enable

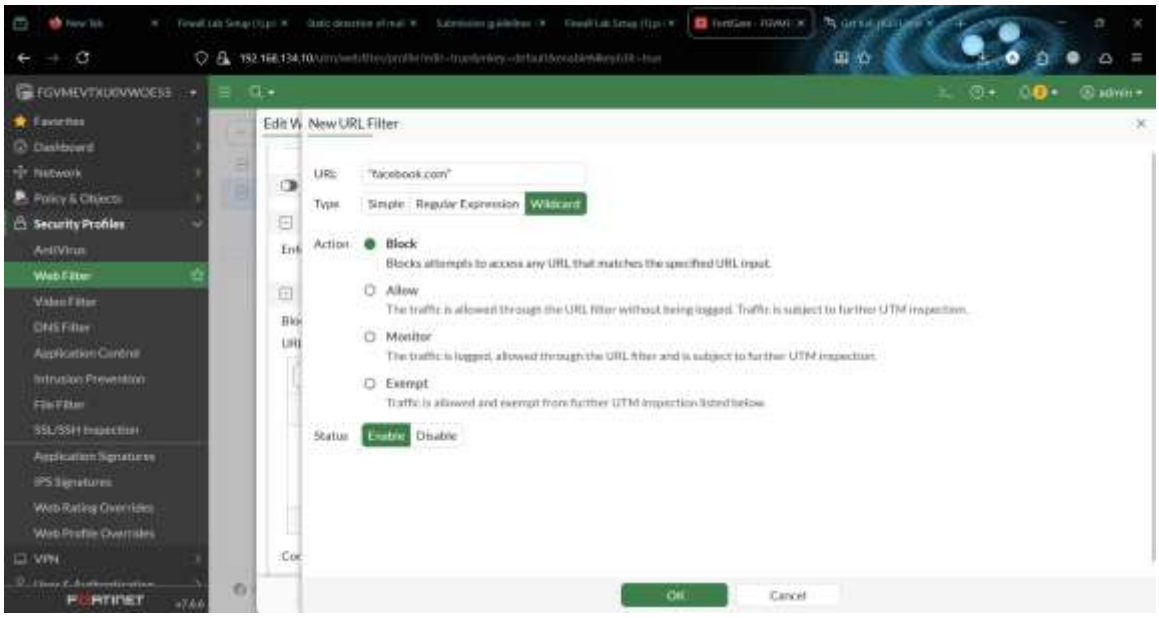


Figure 4.14 — URL filter entry configuration: facebook.com, Wildcard type, Block action, Enabled status

Using the Wildcard type ensures that all subdomains of facebook.com (e.g., m.facebook.com, static.facebook.com, connect.facebook.net) are also blocked, preventing evasion through subdomain access.

4.3.3 Create Web Filter Firewall Policy

A dedicated firewall policy applies the web filter profile to the relevant traffic flow. This policy is named "No-Facebook-Internet-Access" and is placed above the baseline "Internet-Traffic" policy in the policy evaluation order.

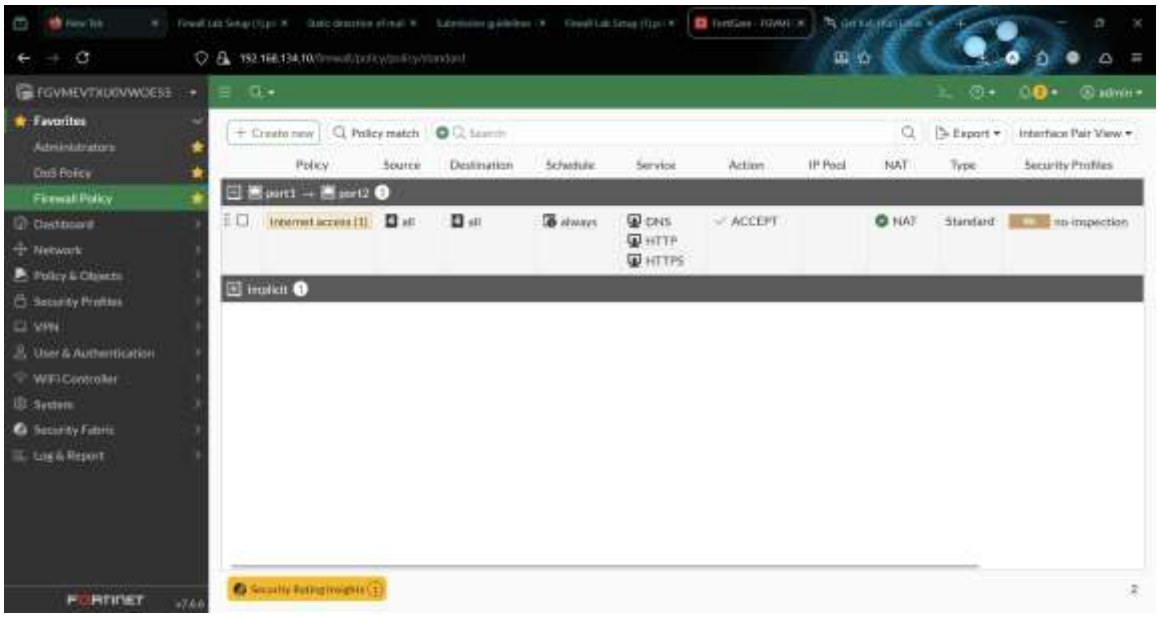


Figure 4.15 — Creating a new firewall policy for the web filter profile application

Policy Parameter	Configured Value
Policy Name	No-Facebook-Internet-Access
Incoming Interface	port3 (LAN / Internal)
Outgoing Interface	port2 (WAN / Internet-facing)
Source	all
Destination	all
Service	ALL
Action	ACCEPT (web filter profile handles blocking at URL level)
NAT	Enabled — Use Outgoing Interface Address
Security Profile — Web Filter	Web filter profile with facebook.com URL block rule applied
Policy Position	TOP of policy list — must evaluate before Internet-Traffic policy

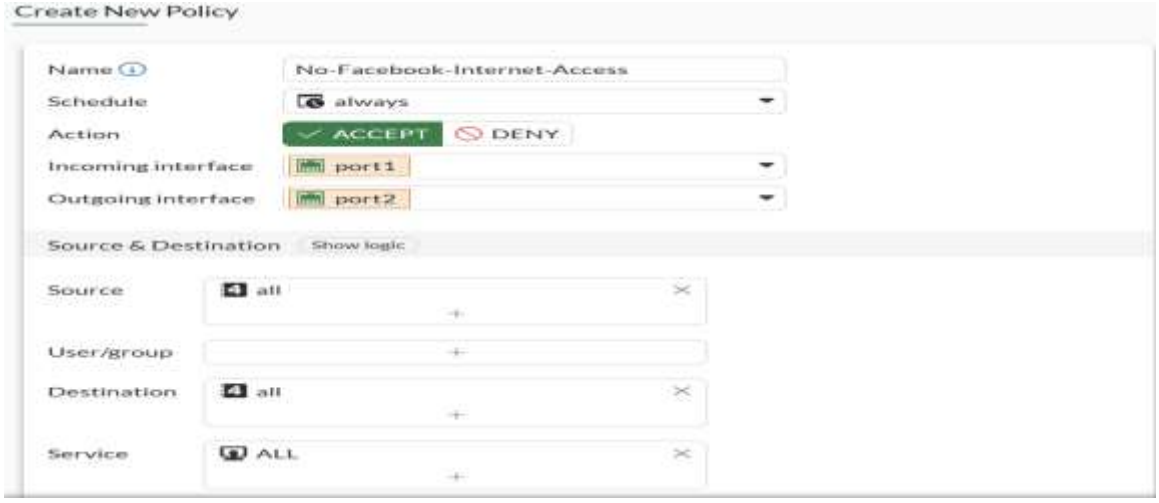


Figure 4.16 — Web filter firewall policy showing interface, source/destination, and security profile assignment

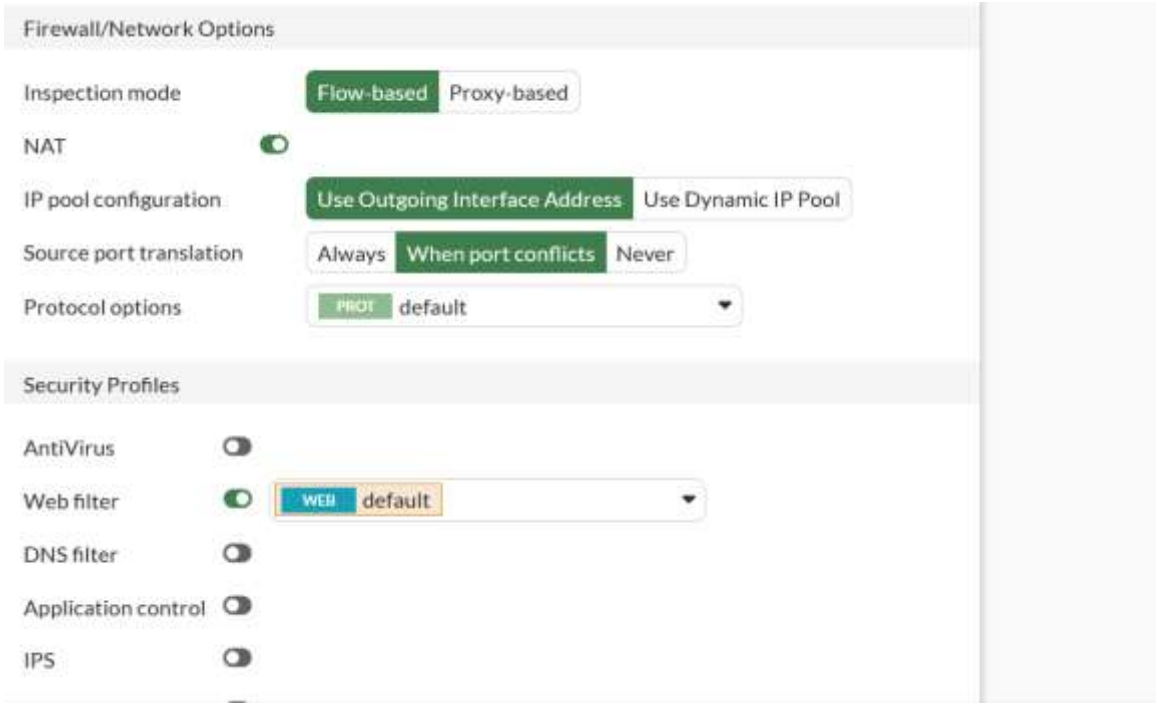


Figure 4.17 — Web filter profile attached to the policy in the Security Profiles section

Figure 4.18 — Policy list "By Sequence" view confirming No-Facebook-Internet-Access is at the top of the evaluation order

Critical Note on Policy Order:
FortiGate evaluates firewall policies top-to-bottom in sequence. The web filter policy MUST appear before the baseline Internet-Traffic policy, otherwise traffic will match the permissive policy first and bypass the URL filter entirely.

5. Task 4 — Demonstration & Validation

This task provides empirical validation of the security policies configured in Task 3. Two distinct test scenarios are executed: URL-based website blocking verification and active DoS attack detection and mitigation testing using Kali Linux as the attacker machine.

5.1 Website Blocking Demonstration

The web content filtering policy is validated by attempting to access facebook.com and its associated domains from a LAN client. A successful block is confirmed by the FortiGate block page appearing in the browser instead of the Facebook interface.

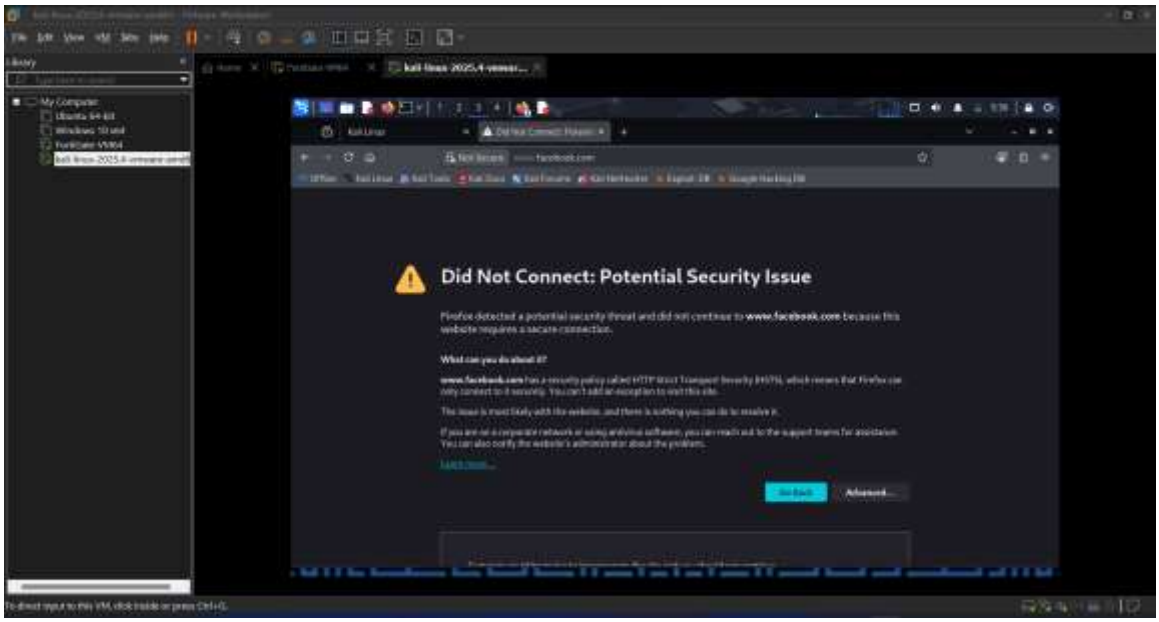
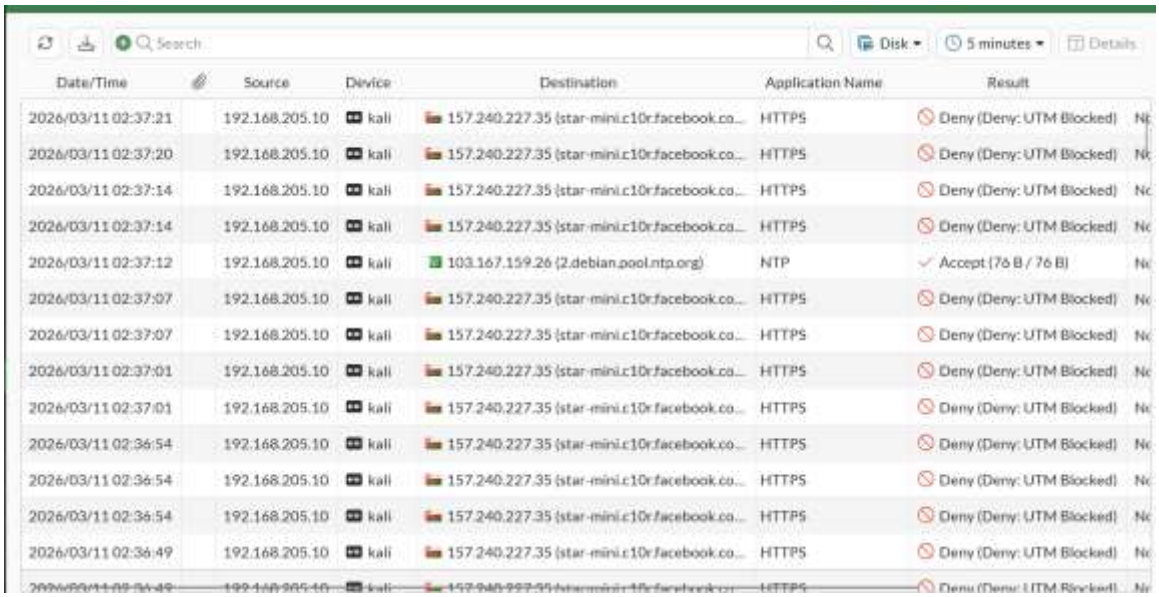


Figure 5.1 — Browser attempting to load facebook.com — FortiGate intercepts the HTTPS request



Date/Time	Source	Device	Destination	Application Name	Result
2026/03/11 02:37:21	192.168.205.10	kali	157.240.227.35 (star-mini.c10r.facebook.co...	HTTPS	Deny (Deny: UTM Blocked)
2026/03/11 02:37:20	192.168.205.10	kali	157.240.227.35 (star-mini.c10r.facebook.co...	HTTPS	Deny (Deny: UTM Blocked)
2026/03/11 02:37:14	192.168.205.10	kali	157.240.227.35 (star-mini.c10r.facebook.co...	HTTPS	Deny (Deny: UTM Blocked)
2026/03/11 02:37:14	192.168.205.10	kali	157.240.227.35 (star-mini.c10r.facebook.co...	HTTPS	Deny (Deny: UTM Blocked)
2026/03/11 02:37:12	192.168.205.10	kali	103.167.159.26 (2.debian.pool.ntp.org)	NTP	Accept (76 B / 76 B)
2026/03/11 02:37:07	192.168.205.10	kali	157.240.227.35 (star-mini.c10r.facebook.co...	HTTPS	Deny (Deny: UTM Blocked)
2026/03/11 02:37:07	192.168.205.10	kali	157.240.227.35 (star-mini.c10r.facebook.co...	HTTPS	Deny (Deny: UTM Blocked)
2026/03/11 02:37:01	192.168.205.10	kali	157.240.227.35 (star-mini.c10r.facebook.co...	HTTPS	Deny (Deny: UTM Blocked)
2026/03/11 02:37:01	192.168.205.10	kali	157.240.227.35 (star-mini.c10r.facebook.co...	HTTPS	Deny (Deny: UTM Blocked)
2026/03/11 02:36:54	192.168.205.10	kali	157.240.227.35 (star-mini.c10r.facebook.co...	HTTPS	Deny (Deny: UTM Blocked)
2026/03/11 02:36:54	192.168.205.10	kali	157.240.227.35 (star-mini.c10r.facebook.co...	HTTPS	Deny (Deny: UTM Blocked)
2026/03/11 02:36:54	192.168.205.10	kali	157.240.227.35 (star-mini.c10r.facebook.co...	HTTPS	Deny (Deny: UTM Blocked)
2026/03/11 02:36:49	192.168.205.10	kali	157.240.227.35 (star-mini.c10r.facebook.co...	HTTPS	Deny (Deny: UTM Blocked)

Figure 5.2 — FortiGate block page displayed for facebook.com, confirming the URL filter policy is active

Deny (Deny: UTM Blocked)	No-Facebook-Internet-Access (3)
Deny (Deny: UTM Blocked)	No-Facebook-Internet-Access (3)
Deny (Deny: UTM Blocked)	No-Facebook-Internet-Access (3)
Deny (Deny: UTM Blocked)	No-Facebook-Internet-Access (3)
Accept (76 B / 76 B)	No-Facebook-Internet-Access (3)
Deny (Deny: UTM Blocked)	No-Facebook-Internet-Access (3)
Deny (Deny: UTM Blocked)	No-Facebook-Internet-Access (3)
Deny (Deny: UTM Blocked)	No-Facebook-Internet-Access (3)
Deny (Deny: UTM Blocked)	No-Facebook-Internet-Access (3)
Deny (Deny: UTM Blocked)	No-Facebook-Internet-Access (3)
Deny (Deny: UTM Blocked)	No-Facebook-Internet-Access (3)
Deny (Deny: UTM Blocked)	No-Facebook-Internet-Access (3)
Deny (Deny: UTM Blocked)	No-Facebook-Internet-Access (3)

Figure 5.3 — Detailed FortiGate block notification showing the policy name and category that triggered the block

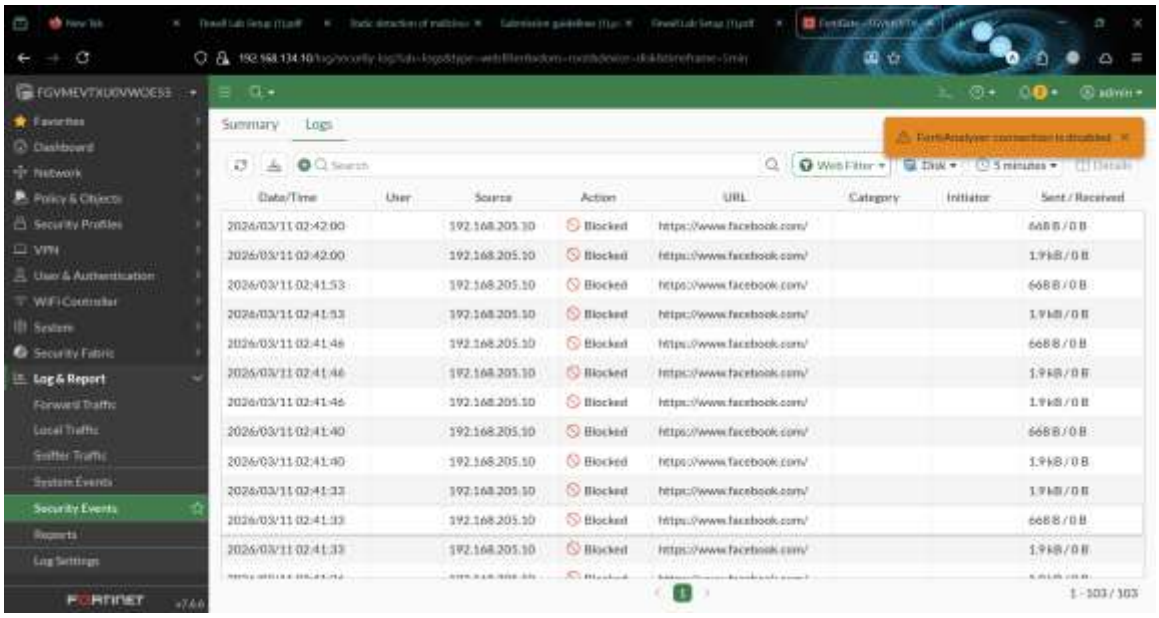


Figure 5.4 — FortiGate web filter log entry showing the blocked facebook.com access attempt with timestamp and source IP

Test Parameter	Result
Target URL	https://www.facebook.com
Expected Outcome	Access denied — FortiGate block page displayed
Actual Outcome	✓ BLOCKED — Block page served by FortiGate
Matching Policy	No-Facebook-Internet-Access (top of policy list)
Block Mechanism	Static URL Filter — Wildcard match on facebook.com

5.2 DoS Attack Detection Demonstration

The IPv4 DoS anomaly protection policy is validated using hping3 on Kali Linux — a widely used network packet crafting tool that can generate high-rate SYN flood attacks. The attack is directed at the FortiGate management IP (192.168.205.2) to test the tcp_syn_flood anomaly profile.

5.2.1 Attack Command

The following hping3 command generates a high-rate TCP SYN flood with randomised source IPs to simulate a distributed DoS (DDoS) attack:

```
hping3 -c 15000 -d 120 -S -w 64 -p 80 --flood --rand-source
192.168.205.2
```

Flag	Description

-c 15000	Send 15,000 packets total
-d 120	Packet data size: 120 bytes
-S	Set TCP SYN flag — simulates connection initiation flood
-w 64	TCP window size: 64 bytes
-p 80	Target port: 80 (HTTP)
--flood	Send packets as fast as possible without waiting for replies
--rand-source	Randomise source IP addresses — simulates spoofed DDoS traffic

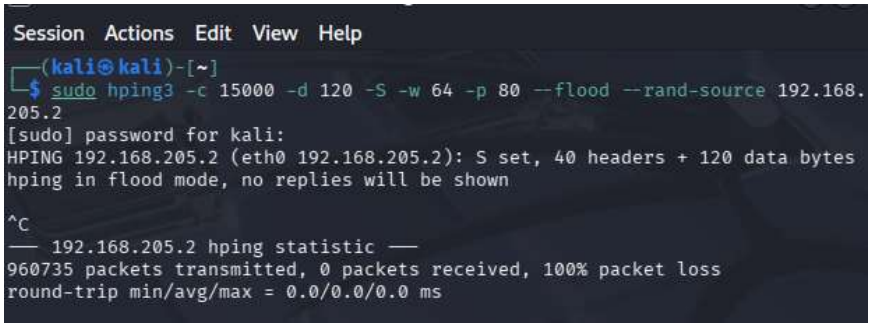


Figure 5.5 — hping3 SYN flood command executed from Kali Linux targeting FortiGate management IP

The FortiGate DoS protection policy detects the SYN flood within milliseconds of the threshold being crossed and begins dropping the flood traffic. The Kali Linux terminal shows no received responses confirming that all SYN packets are being dropped.

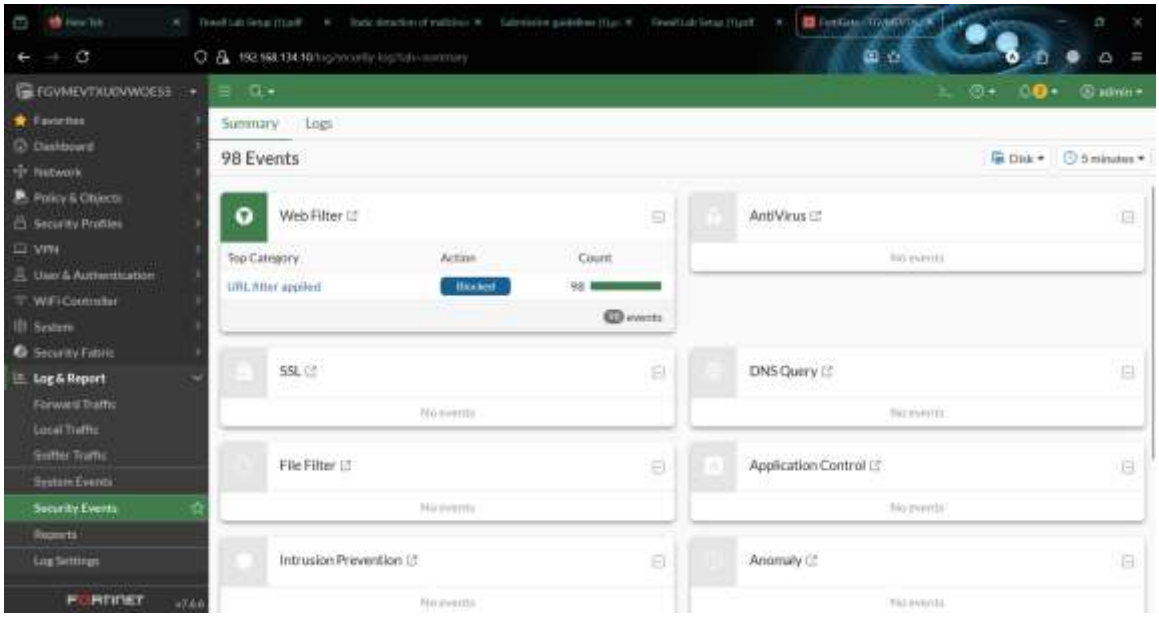


Figure 5.6 — FortiGate DoS policy logs showing tcp_syn_flood anomaly triggered and packets blocked

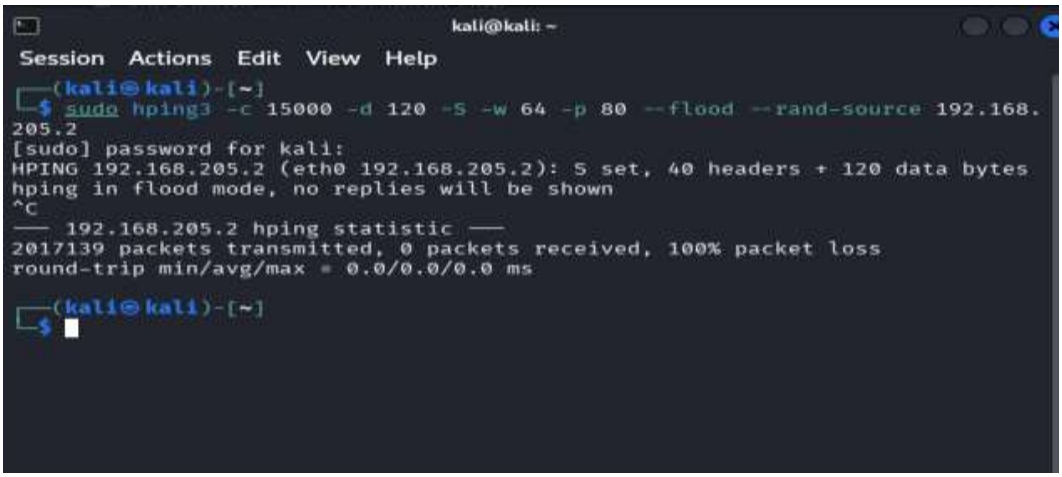


Figure 5.7 — FortiGate security event log detail confirming DoS anomaly detection with source, destination, and threshold data

Test Parameter	Result
Attack Type	TCP SYN Flood with randomised source IPs
Attack Tool	hping3 on Kali Linux
Target	192.168.205.2 (FortiGate management IP), port 80
Packet Count	15,000 SYN packets at maximum flood rate
Expected Outcome	DoS policy triggers — flood traffic blocked
Actual Outcome	✓ BLOCKED — All flood traffic dropped by DoS anomaly policy
Anomaly Triggered	tcp_syn_flood — threshold exceeded
Log Generated	Security event log entry created with source, destination, and anomaly details

6. Conclusions & Recommendations

6.1 Summary of Findings

This laboratory exercise successfully demonstrated the deployment and operational configuration of a FortiGate Next-Generation Firewall in a virtualised enterprise environment. All four tasks were completed and validated:

- FortiGate VM was successfully deployed on VMware Workstation with correct virtual NIC-to-interface mapping across six network segments
- Management interface security was verified: HTTPS-only access confirmed, admin credentials hardened, and authenticated GUI access tested
- Three complementary security policies were implemented: baseline IPv4 firewall policy with NAT, DoS anomaly protection covering 15 L3/L4 attack profiles, and URL-based web content filtering
- Policy effectiveness was empirically validated through browser-based URL access testing (Facebook blocked) and active SYN flood simulation using hping3 (attack detected and blocked)

6.2 Security Recommendations for Production Deployment

While the lab configuration demonstrates functional FortiGate capabilities, a production deployment would require additional hardening measures:

Recommendation	Rationale
Enable FortiToken 2FA	Prevents admin account compromise even if credentials are stolen
Restrict Trusted Hosts	Limits GUI/SSH access to specific management workstation IPs
Enable SSL/TLS Inspection	Required to inspect HTTPS traffic for malware and data exfiltration
Enable IPS Signatures	Provides exploit and vulnerability-based detection beyond anomaly thresholds
Configure Syslog / SIEM	Forward logs to a central SIEM for correlation and long-term retention
Regular Firmware Updates	Apply FortiOS security patches to remediate known CVEs
Geo-IP Blocking	Block inbound connections from high-risk geographic regions
Tune DoS Thresholds	Calibrate thresholds based on baseline traffic profiling to minimise false positives

6.3 Knowledge Outcomes

Upon completion of this laboratory assignment, the following competencies have been demonstrated:

- Virtualised NGFW deployment and lifecycle management using FortiGate on VMware
- CLI-based interface configuration and administrative access control
- Zone-based network segmentation with WAN/LAN/Management separation
- Stateful firewall policy construction with NAT for outbound internet access
- Layer 3/4 DoS anomaly detection and threshold-based blocking policy configuration
- Application-layer web content filtering with URL pattern matching
- Security policy validation methodologies using penetration testing tools (hping3)
- Evidence-based documentation of security controls for audit and compliance purposes

7. References

The following references were consulted in the preparation of this laboratory report:

- Fortinet, Inc. (2024). FortiOS Administration Guide — Firewall Policies. Fortinet Documentation Library. https://docs.fortinet.com/document/fortigate
- Fortinet, Inc. (2024). FortiGate VM Installation Guide for VMware. Fortinet Documentation Library. https://docs.fortinet.com/document/fortigate-vm
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